Artificial Intelligence Review (2025) 58:86
https://doi.org/10.1007/s10462-024-11063-z

A comprehensive survey on impact of applying various technologies on the internet of medical things

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Accepted: 30 November 2024 / Published online: 8 January 2025

Abstract

This paper explores the transformative impact of the Internet of Medical Things (IoMT) on healthcare. By integrating medical equipment and sensors with the internet, IoMT enables real-time monitoring of patient health, remote patient care, and individualized treatment plans. IoMT significantly improves several healthcare domains, including managing chronic diseases, patient safety, and drug adherence, resulting in better patient outcomes and reduced expenses. Technologies like blockchain, Artificial Intelligence (AI), and cloud computing further boost IoMT's capabilities in healthcare. Blockchain enhances data security and interoperability, AI analyzes massive volumes of health data to find patterns and make predictions, and cloud computing offers scalable and cost-effective data processing and storage. Therefore, this paper provides a comprehensive review of the Internet of Things (IoT) and IoMT-based edge-intelligent smart healthcare, focusing on publications published between 2018 and 2024. The review addresses numerous studies on IoT, IoMT, AI, edge and cloud computing, security, Deep Learning, and blockchain. The obstacles facing IoMT are also covered in this paper, including interoperability issues, regulatory compliance, and privacy and data security concerns. Finally, recommendations for further studies are provided.

Keywords Internet of Medical Things (IoMT) · Smart healthcare · Deep Learning (DL) · Blockchain · Artificial Intelligence (AI) · Cloud computing

Abbreviation

AI	Artificial intelligence
CNN	Convolutional neural network
DDoS	Distributed denial-of-service
EHRs	Electronic health records
EEGs	Electroencephalograms
GDPR	General data protection regulation
IoT	Internet of things
IaaS	Infrastructure as a service
LR	Logistic regression

Extended author information available on the last page of the article

NLP	Natural language processing
NLG	Natural language generation
PRISMA	Preferred reporting items for systematic reviews and meta-analyses
RF	Random forest
SaaS	Software as a service
BIoMT	Blockchain-based IoMT
DL	Deep Learning
DoS	Denial-of-service
ECGs	Electrocardiograms
EMGs	Electromyograms
HIPAA	U.S. health insurance portability and accountability act
IoMT	Internet of medical things
ILSVRC	ImageNet large scale visual recognition competition
ML	Machine learning
NER	Named entity recognition
NNs	Neural networks
PaaS	Platform as a service
SVM	Support vector machine

1 Introduction

The IoMT is revolutionizing the healthcare sector by enabling a system of medical equipment and software that connects to healthcare information technology systems through computer networks over the internet to gather and transmit patient data, track patient health, and provide healthcare professionals with real-time information (Din et al. 2019; Qureshi and Krishnan 2018). IoMT technology includes wearable biosensors, implanted technology, mobile health applications, and remote monitoring systems (Ashfaq et al. 2022), which can be used to manage chronic illnesses, track medication adherence, and monitor vital signs (Ashfaq et al. 2022; Mehrdad et al. 2021). The data collected by IoMT devices can be safely communicated to healthcare professionals and connected with Electronic Health Records (EHRs) to enable more individualized and proactive healthcare delivery (Qureshi and Krishnan 2018). IoMT technology has the potential to transform healthcare delivery by providing healthcare practitioners with real-time patient data and insights, helping to increase patient engagement, lower healthcare expenses, and improve patient outcomes and care quality (Gahlot et al. 2018; Chaudhry et al. 2021). Personalized, effective, and accessible healthcare is one of the main benefits of IoMT (Vishnu et al. 2020; Bhushan et al. 2023).

The potential impact of IoMT on the healthcare sector is enormous as digital technology continues to be adopted (Phan et al. 2022; Wang and Song 2023), and it is rapidly becoming a crucial aspect of contemporary healthcare. Wearable technology, mobile health applications, and remote monitoring systems have transformed how people and healthcare professionals communicate (Ben Dhaou et al. 2021; Da Costa et al. 2018). Wearable biosensors and implantable devices can monitor vital signs such as heart rate, blood pressure, and glucose levels, and the data can be sent in real-time to healthcare professionals (Phan et al. 2022; Chakraborty et al. 2020). As a result, medical professionals can make more informed decisions regarding patient care and take action before a problem worsens (Albahri et al. 2019; El Khatib et al. 2023).

IoMT effectively impacts both patients and healthcare personnel. For patients, IoMT can assist them in developing a better understanding of their health and making informed decisions about their care, which can increase patient engagement and improve patient outcomes (Ben Dhaou et al. 2021; Srivastava and Routray 2022). IoMT can also potentially reduce healthcare expenses by enabling remote patient monitoring technologies to monitor patients at home, reducing hospital stays and readmissions (Chopade et al. 2023). Additionally, researchers can use IoMT to gather a large amount of patient data, which could help accelerate the pace of medical innovation and improve the standard of patient care (Alam et al. 2018; Villegas-Ch et al. 2023). In general, the potential of IoMT to significantly enhance the quality of healthcare can be summarized in the following effects (Chen et al. 2021; Albahri et al. 2023):

- **Adequate patient outcomes**: The IoMT enables healthcare professionals to remotely monitor patients and take early action if they identify any issues. This could improve patient outcomes, leading to fewer hospital stays and reduced medical expenses (Pal et al. 2018). IoMT has the potential to significantly improve patient outcomes in various ways, including (Nishad and Tripathi 2020; Gupta et al. 2023):
  - **Continuous monitoring**: IoMT devices can continuously monitor vital signs such as heart rate, blood pressure, and glucose levels, which allows for early detection of potential health issues and timely intervention.
  - **Chronic disease management**: Patients with chronic conditions, such as diabetes or heart disease, can be monitored remotely, reducing the need for frequent hospital visits and enabling more consistent management of their conditions.
  - **Telehealth services**: IoMT facilitates telehealth consultations, allowing patients to receive medical advice and care without the need to travel to healthcare facilities.
  - **Predictive analytics**: Based on Machine Learning (ML) algorithms, IoMT can predict health events such as heart attacks or strokes before they occur, enabling preventive measures to be taken.
  - **Smart medication management**: IoMT devices can remind patients to take their medications on time and track adherence.
  - **Automated alerts**: Healthcare providers can receive alerts if patients fail to take their medication, allowing for prompt follow-up and support.
  - **Workflow optimization**: By monitoring patient flow and resource utilization, IoMT can help optimize hospital workflows, reducing wait times and improving the overall patient experience.
  - **Health education**: Connected devices can provide patients with educational content and reminders about healthy lifestyle choices.
  - **Preventive care**: By enabling early detection and continuous monitoring, IoMT can reduce the need for expensive emergency care and hospitalizations.
  - **Efficient resource use**: IoMT helps in optimizing the use of healthcare resources, reducing waste, and lowering operational costs.
  - **Integrated care**: IoMT facilitates better communication and data sharing among healthcare providers, leading to more coordinated and comprehensive patient care.

- **Real-time data access**: Clinicians can access real-time patient data from anywhere, allowing for timely decision-making and interventions.

- **Enhance patient engagement**: With the help of IoMT devices, patients can monitor their health and receive personalized feedback and assistance. Increased patient involvement and motivation to adhere to treatment recommendations may lead to better health outcomes (Khan et al. 2023). IoMT can enhance patient engagement in several ways, including (Dwivedi et al. 2022; Mishra and Singh 2023; Roy and Nahid 2022):

  - **Personalized health data and insights**: Using wearables like fitness trackers and smartwatches to monitor health metrics (e.g., heart rate, steps, sleep patterns). Provide personalized feedback and recommendations based on this data. Developing apps that aggregate data from various IoMT devices and present it in an easily understandable format.
  - **Interactive and user-friendly platforms**: Designing user-friendly interfaces for IoMT devices and associated apps, making it easy for patients to navigate and understand their health data.
  - **Education and training**: Providing educational resources on how to use IoMT devices effectively and take appropriate actions based on insights. Offering online tutorials, webinars, and customer support to assist patients in understanding and utilizing their IoMT devices.
  - **Two-way communication**: Enabling healthcare providers to monitor patients' health data remotely and communicate directly with patients. Allowing patients to provide feedback on their experiences, report issues, and suggest improvements.
  - **Privacy and security**: Implementing robust data security measures to protect patient information and build trust in IoMT technologies. Clearly communicating data privacy policies and obtaining informed consent from patients regarding using their health data.
  - **Continuous improvement**: Regularly collecting and analyzing patient feedback to continuously improve IoMT devices and services. Staying abreast of technological advancements and incorporating innovative solutions to enhance patient engagement and health outcomes.
  - **Incentives and rewards**: Offering incentives such as discounts or rewards for consistently using IoMT devices and active engagement with health management programs. Partnering with health insurers to offer premium discounts for patients who actively engage with IoMT devices and maintain healthy behaviors.

- **Rapid and accurate diagnosis**: Real-time data from IoMT devices can be used to detect health problems early, leading to more reasonable and effective treatment regimens (Wang and Song 2023).
- **Efficiency gains**: IoMT can automate repetitive processes, freeing up healthcare professionals to concentrate on higher-level decision-making. This could result in more effective resource allocation and higher efficiency (Pal et al. 2018; Shanmugam and Azam 2023).
- **Better care coordination and improved patient outcomes**: IoMT can facilitate more differentiated communication and collaboration among healthcare practitioners, leading

to improved patient outcomes (Sutradhar et al. 2023).

IoMT offers numerous benefits for healthcare personnel, significantly reducing their workload through automatic data collection and real-time alerts. These benefits can be summarized as follows (Mitra et al. 2022; El-Saleh et al. 2024):

- Continuous monitoring of a patient's vitals and other health indicators reduce the need for frequent manual checks by healthcare personnel.
- Automatic collection and transmission of health data to EHRs minimize the time and effort required for manual data entry.
- Real-time detection of critical changes in a patient's condition and alerts to healthcare providers enable prompt intervention and reduce time spent on routine monitoring.
- Healthcare personnel can allocate more time to direct patient care and other critical tasks, improving overall workflow efficiency.
- The risk of human error is reduced, ensuring accurate and comprehensive data. This leads to better-informed clinical decisions and improved patient outcomes.
- Administrative tasks such as documentation and data analysis are simplified, allowing healthcare personnel to spend more time caring for patients.
- Automated data collection and real-time alerts optimize resource allocation in healthcare facilities, ensuring staff are deployed where they are most needed and reducing unnecessary workload.
- Empowered patients can manage their health more effectively, reducing the frequency of consultations and the demand on healthcare personnel.

On the other hand, the IoMT considers the key factor in the creation and development of smart hospitals, which are hospitals that employ technology to enhance patient outcomes, reduce expenses, and streamline operations (Chaudhry et al. 2021). The adoption of IoMT in smart hospitals can revolutionize the medical field. The following IoMT features are present in smart hospitals (Haleem et al. 2023):

- **Remote patient monitoring**: IoMT devices enable real-time monitoring of patient health outside the hospital (Chopade et al. 2023). Wearable biosensors and medical monitors gather and transmit vital signs, such as heart rate, blood pressure, and oxygen levels, to healthcare specialists (Shreya et al. 2022; Panja et al. 2023; Liu et al. 2023). This allows medical professionals to monitor patients remotely and intervene as needed, reducing the need for hospital readmissions and improving patient outcomes (Zhang et al. 2018; Chen et al. 2023).
- **Predictive medical equipment maintenance**: By collecting and analyzing data related to the use and operation of medical equipment, IoMT devices can help hospitals anticipate maintenance needs. This helps them avoid expensive repairs, reduce equipment downtime, improve patient safety, and save costs (Monteiro et al. 2023; Srivastava and Routray 2022).
- **Streamlined operations**: IoMT can streamline hospital operations by automating certain procedures. For example, IoMT devices can track medical supplies, automate inventory management, and place new supply orders as needed (Shreya et al. 2022; Singh et al. 2022; He et al. 2022). IoMT devices can enhance patient flow by monitoring

patient movement and speeding up patient discharge times. This reduces both time and administrative costs (Zhang et al. 2018; Alkatheiri and Alghamdi 2023).

- **Real-time patient data collection**: IoMT devices collect real-time patient data, enabling the development of individualized treatment regimens based on each patient's unique health information (Jain et al. 2021). By analyzing patient data, healthcare professionals can better meet each patient's needs, leading to lower healthcare expenditures and improved patient outcomes (Dwivedi et al. 2022; Panja et al. 2023; Webber et al. 2022; Taami et al. 2023).

Real-time patient data is gathered by sensors and connected devices in an IoT-based healthcare system. To be processed and stored, this data is sent to a central node (Monteiro et al. 2023). Remote data access allows healthcare practitioners to make well-informed decisions (Chakraborty et al. 2020; Shreya et al. 2022). Advanced analytics and ML gather insightful data that can be used for preventative interventions and individualized care (Monteiro et al. 2023). The effectiveness of healthcare, as well as patient outcomes, are improved by this method (Shreya et al. 2022), as shown in Fig. 1.

Moreover, wearable biosensors, which are small electronic devices, can be employed in the healthcare sector to enhance patient care, promote medical research, and reduce healthcare costs by using the IoMT (Ben Dhaou et al. 2021). These biosensors are frequently incorporated into wearable gadgets, such as smartwatches, fitness trackers, and other types of clothing or accessories, that can connect to technologies like smartphones or computers (Mehrdad et al. 2021; Hurley and Popescu 2021). Wearable biosensors can track and collect various data types, including movement, heart rate, temperature, and other physiological indicators (Malamas et al. 2021; Rajput and Brahimi 2019). They can be used for various purposes, such as tracking athletic progress, improving athletes' performance, conducting medical research, providing healthcare, and delivering higher-quality treatment remotely. Here are some applications of wearable biosensors in healthcare that make use of the IoMT (Haleem et al. 2023):

- Allowing patients with chronic diseases, such as diabetes or high blood pressure, to monitor their health and receive individualized feedback and suggestions from healthcare specialists (Devi et al. 2023).
- Tracking a patient's vital signs, including heart rate, blood pressure, and respiration rate, and they can provide real-time data to medical experts. This allows medical experts to observe patients remotely and identify early indications of illnesses like heart failure or

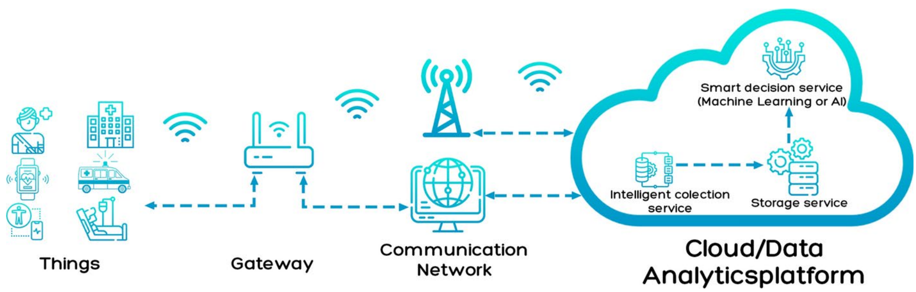

Fig. 1 An overview of a typical IoT-based healthcare system

respiratory distress (Malamas et al. 2021; Khan et al. 2023).
- Gathering factual information about patients' outcomes in clinical trials, such as drug efficacy or adverse events. Therefore, fewer in-person meetings may be required, and data collection might become more accurate and efficient (Sinha et al. 2023).
- Monitoring the health and safety of elderly people by spotting falls or tracking medication compliance, for example. This can improve quality of life and reduce hospitalization requirements (Gahlot et al. 2018).
- Detecting changes in heart rate or respiration rate, which can offer crucial information in emergency situations. That can facilitate healthcare providers to respond quickly and provide proper care (Malamas et al. 2021).

Furthermore, IoMT offers numerous cost-reduction strategies that can improve the efficiency of healthcare systems, reduce costs, and simplify processes. Here's an explanation of how to implement cost-reduction strategies in IoMT (Arora 2020; Nishad and Tripathi 2020):

- **Optimizing Device Utilization**

  - Utilize platforms for maintenance planning and real-time monitoring. Predictive analytics can be employed to anticipate equipment breakdowns and schedule maintenance only when necessary.
  - Share devices among departments to optimize usage and avoid unnecessary expenditures. Algorithms can be used to determine the most required devices based on patient needs and device availability.

- **Reducing Operational Costs**

  - Leverage cloud computing services to process, store, and analyze data. Utilize usage-based cloud service models to further reduce upfront costs by paying only for services used.
  - Integrate IoMT devices with EHR systems to improve data collection and reduce human entry errors.

- **Enhancing Clinical Efficiency**

  - Reduce the need for in-person visits by using IoMT devices for remote patient monitoring. Early detection of health issues through IoMT can lead to timely interventions and reduce long-term treatment costs.
  - Utilize analytics tools to analyze data from IoMT devices for insights that can improve patient outcomes and operational efficiencies. Data can be used to measure treatment effectiveness and adjust practices for better patient care and cost reduction.

- **Strengthening Security**

  - Implement robust cybersecurity measures to protect patient data and prevent costly breaches. Ensure compliance with regulations to avoid fines and legal expenses.
  - Regularly assess the security of IoMT devices and networks to identify and address vulnerabilities.

- **Streamlining Administrative Processes**

  - Implement tools to automate routine administrative tasks such as data entry, reporting, and scheduling.
  - Re-engineer workflows to eliminate inefficiencies and streamline processes.

- **Training and Support**

  - Provide comprehensive training for staff to ensure effective use of IoMT devices and systems.
  - Offer ongoing support to address issues quickly and minimize downtime.

- **Improving Patient Outcomes**

  - Educate patients about their health conditions and the use of IoMT devices to improve adherence to treatment plans.
  - Implement feedback systems to gather patient and provider input for continuous improvement.

This study provides a comprehensive examination of the influence of the IoMT on healthcare systems, focusing on the years 2018 to 2023 (Khan et al. 2023; Shakeel et al. 2023). IoMT enables real-time patient health monitoring, remote patient monitoring, and individualized treatment regimens by linking medical devices and sensors to the Internet. Our study highlights the significant advancements IoMT has already achieved in managing chronic diseases, improving patient safety, and enhancing drug adherence (Si-Ahmed et al. 2023). To further increase the potential impact of IoMT in healthcare, we also investigate the integration of cutting-edge technologies such as blockchain, artificial intelligence (AI), and cloud computing. Cloud computing ensures affordable and scalable processing and storage of health data, while blockchain improves data security and interoperability. AI algorithms can also evaluate vast amounts of health data for patterns and forecasts (Alhaj et al. 2022; Shakeel et al. 2023).

Our study takes a more comprehensive perspective by examining how IoT and IoMT work together in smart healthcare, in contrast to earlier studies that may have focused on specific IoMT or IoT features (Khan et al. 2023). Analyzing journal articles from 2018 to 2023, we ensure that our study considers the most recent advancements and trends in this rapidly evolving field (Shakeel et al. 2023). Our research also addresses the challenges of implementing IoMT, including security, DL, and blockchain integration (Alhaj et al. 2022). We aim to increase understanding of the barriers that must be overcome for successful IoMT adoption by providing insights into these challenges (Si-Ahmed et al. 2023).

Moreover, our work provides valuable suggestions for further study, going beyond a simple literature review. We recognize that the field of IoMT is rapidly evolving, and there are still many opportunities for research and development (Shakeel et al. 2023; Alhaj et al. 2022). By highlighting crucial topics for further study, we hope to inspire researchers and healthcare professionals to explore novel approaches and advancements in IoMT technology (Si-Ahmed et al. 2023). Our research will ultimately contribute to the advancement of IoMT in healthcare systems, ensuring its continued positive impact on patient outcomes and healthcare effectiveness (Khan et al. 2023).

Healthcare is a fundamental aspect of life, yet modern healthcare systems are increasingly strained due to a steadily aging population and a corresponding rise in chronic illnesses. This escalating demand for resources, including hospital beds, doctors, and nurses, highlights the urgent need for solutions to alleviate pressure on these systems while maintaining high-quality care for at-risk patients. In recent years, the healthcare industry has recognized a growing interest in developing effective and practical medical care solutions. These innovations aim to assist hospitals and healthcare providers in streamlining diagnoses and enhancing patient care (Baker et al. 2017; Tunc et al. 2021).

Despite advancements in healthcare technology, a significant gap exists in comprehensive and systematic reviews that consolidate these developments, particularly in the context of the Internet of Things (IoT). The integration of IoT in healthcare holds the potential to revolutionize patient monitoring, data collection, and overall care efficiency. However, there is a paucity of research that systematically examines the scope, implementation challenges, and outcomes of IoT applications in this field. This study aims to address this gap by providing a detailed systematic review of current IoT applications in healthcare, identifying the challenges and benefits, and offering insights into future directions. By focusing on this research gap, our study highlights the novelty and significance of systematically analyzing IoT's role in modern healthcare, thereby contributing valuable knowledge to both academic and practical domains (Baker et al. 2017; Tunc et al. 2021).

1.1 Research methods and strategy

To ensure a focused and rigorous review, we employed the systematic review methodology, as illustrated in Fig. 2. The review process consisted of three sequential stages: identification, screening, and eligibility testing. In the identification phase, we located relevant papers using searches on IEEExplore, Google Scholar, PubMed, and Elsevier data sources, which yielded 168 papers. Through the screening process, we eliminated duplicate and nonconforming documents, resulting in the selection of 153 papers. During the eligibility testing stage, we further refined our selection by eliminating documents unrelated to healthcare, ultimately selecting 140 papers for inclusion in our survey. Table 1 provides a summary of the filtering process for the collected research papers (Rajput and Brahimi 2019).

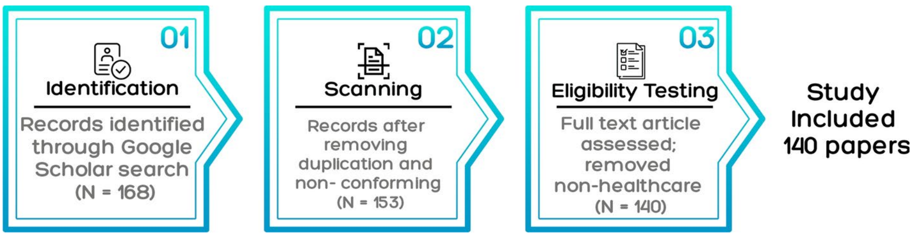

Fig. 2 Study selection diagram for PRISMA

A search strategy was devised to analyze the selected articles, which involved selecting appropriate data sources and using a combination of keywords relevant to IoMT and its applications in smart healthcare, including “IoMT”, “smart healthcare”, “Deep Learning (DL)”, “blockchain”, “security”, and “cloud computing”. These keywords were selected to cover various aspects, such as technologies, methodologies, and specific applications within IoMT. The key phrases “IoMT” and (smart healthcare, DL, security, blockchain, or cloud computing) were combined in different configurations to retrieve articles relevant to the study.

Data sources such as IEEExplore, Google Scholar, PubMed, and Elsevier were chosen due to their extensive archives and comprehensive coverage of relevant technical and medical articles. IEEExplore was chosen for its robust collection of engineering and technology-focused research, which is crucial for understanding the technological aspects of IoMT and their applications. Google Scholar was selected because it provides a broad range of academic articles, offering a wide perspective on IoMT applications across different disciplines. PubMed was included for its comprehensive repository of biomedical and healthcare-related literature, making it a key resource for finding research on the medical applications of IoMT. Elsevier covers various disciplines, including engineering, computer science, healthcare, and biomedical sciences. This multidisciplinary coverage is essential for IoMT research, which intersects technology and healthcare. Elsevier hosts thousands of journals that publish research articles relevant to IoMT, ensuring a comprehensive literature search.

The research areas used to select papers were: “IoMT for smart healthcare” and “Techniques that combined with IoMT for smart healthcare”. Inclusion criteria were studies published between 2018 and 2023 to capture recent advancements, focusing on publications with topics related to IoMT and related technologies in the healthcare sector. Exclusion criteria were set to remove studies that were not relevant to healthcare applications of IoMT or low-quality studies, duplicated publications across different data sources to avoid redundancy, publications that did not conform to the healthcare context of IoMT, and publications without peer review and opinion articles to maintain the quality of the review. This systematic approach ensured a thorough review of the current landscape, comprehensively analyzing IoMT-based smart healthcare, addressing current research challenges, and offering recommendations for future research to optimize IoMT’s effectiveness in healthcare. After removing duplicate and irrelevant content, our search was narrowed down to 140 articles, as shown in Fig. 3.

Table 1 Summary of filtration process of papers found

Keyword	IEEExplore	Google Scholar	PubMed	Elsevier
IOMT	1192	2730	412	1001
Smart healthcare	95	788	444	461
DL	270	1500	83	460
Security	83	1910	132	554
Blockchain	18	1140	56	288
Cloud computing	71	1830	41	538
Selected	50	90	10	10

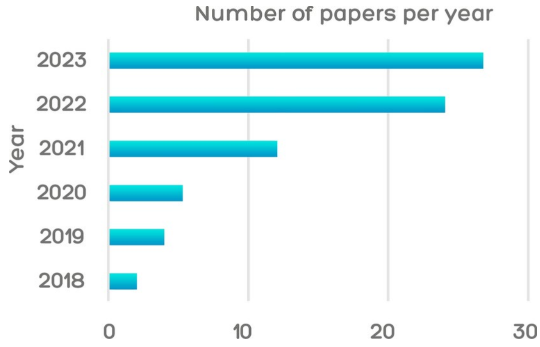

Fig. 3 Number of papers by year

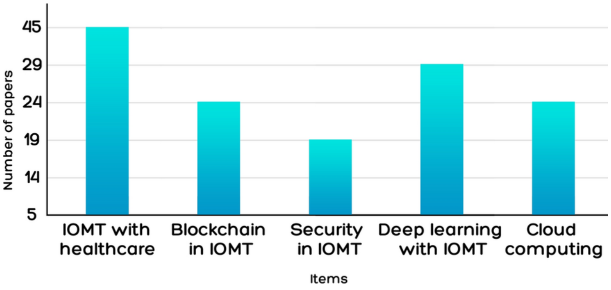

Fig. 4 Number of papers by item

1.2 Data extraction

The following data categories were collected from articles, as shown in Fig. 4:

- Internet of Things (IoT)/IoMT with healthcare
- DL with IoMT
- Blockchain
- Security
- Cloud computing

1.3 Main contributions

The main contributions of this study can be stated as follows:

- **Comprehensive examination of IoMT impact**: The study provides an in-depth analysis of IoMT's influence on healthcare systems, focusing on advancements from 2018 to 2024, including real-time monitoring and personalized treatment plans.
- **Advancements in chronic disease management**: The study highlights significant progress in managing chronic diseases and improving medication adherence through IoMT technologies, integrating AI, blockchain, and cloud computing.
- **Holistic perspective on IoMT in smart healthcare**: The study examines the combined applications of IoMT in smart healthcare, capturing the latest trends and developments comprehensively.
- **Identification of implementation challenges**: The study addresses various challeng-

es in implementing IoMT, including security concerns and the integration of DL and blockchain technology, offering potential solutions.
- **Future research directions**: The study suggests valuable areas for further research to advance IoMT technology, aiming to inspire novel approaches and continued development in the field.
- **Addressing the growing demand in healthcare**: The study underscores the need for effective medical care solutions to alleviate pressure on healthcare systems caused by an aging population and rising chronic illnesses.
- **Systematic review methodology**: Employing a systematic review methodology, the study ensures a focused and rigorous analysis of IoMT applications, challenges, and future directions.

1.4 Structure

The remainder of the paper is formulated as follows. Section 2 presents DL methods with IoMT. The Blockchain-based IoMT (BIoMT) is presented in Sect. 3. Section 4 shows the technologies of cloud-enabled IoMT. Security properties and types of attacks for IoMT are drawn in Sect. 5. Challenges, limitations, and future development are discussed in Sect. 6. Finally, the conclusion is shown in Sect. 7.

2 DL with IoMT

The IoMT uses interconnected sensors, medical equipment, and other technology to track patient health and provide individualized medical care. DL techniques can be used with IoMT to analyze the massive amounts of data produced by various gadgets and sensors and offer insights into patient health, the course of their diseases, and the effectiveness of their treatments. DL models are highly recommended for IoMT applications due to the following reasons:

- **High accuracy and performance**:
  - **Complex data handling**: DL models, especially Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), are highly effective in processing complex medical data such as images, signals, and large-scale EHRs. They have shown superior performance in tasks such as image classification, segmentation, and time-series prediction, which are critical in medical diagnostics and monitoring.
  - **Feature extraction**: DL models automatically extract and learn hierarchical features from raw data, reducing the need for manual feature engineering. This capability is particularly beneficial in medical imaging and biosignal processing, where relevant features may be intricate and difficult to define manually.
- **Versatility and flexibility**:

  - **Multi-modal data integration**: DL can seamlessly integrate various types of medical data, such as images, sensor signals, and textual records, providing a comprehensive analysis and improving diagnostic accuracy.
  - **Adaptability**: These models can be fine-tuned and adapted to specific medical applications and datasets, ensuring that they can cater to a wide range of IoMT applications, from chronic disease management to acute care.

- **Real-time processing**:

  - **Edge computing**: With advancements in edge computing, deploying lightweight versions of DL models on IoMT devices enables real-time data processing and decision-making. This capability is crucial for applications such as continuous patient monitoring and emergency response.

However, in specific scenarios such as resource-constrained environments, the need for interpretability, or when dealing with small datasets, alternative techniques like traditional ML models or simpler neural networks should be considered. This balanced approach ensures that the most appropriate and effective techniques are utilized based on the specific requirements and constraints of each IoMT application. DL algorithms can be utilized in a wide range of IoMT applications (Manimurugan et al. 2020), including predictive analytics, image, and signal processing, the analysis of EHRs and other medical documents to extract pertinent data that will enhance clinical decision-making (Radoglou-Grammatikis et al. 2021), the development of personalized treatment recommendations based on patient health data, medical history, and genetic data (Al-Sa'D et al. 2018), and monitoring the performance of medical devices in real-time, which helps detect any problems and notify healthcare experts (Amin et al. 2019). The subsections that follow provide a succinct explanation of some of these IoMT applications that use DL techniques:

2.1 Predictive analytics

Predictive analytics is a sub-field of data science that studies past data and forecasts future occurrences or outcomes relying on statistical algorithms, ML, and AI (Manickam et al. 2022). It can significantly improve patient outcomes and lower healthcare expenses in the context of the IoMT by identifying possible health concerns, forecasting the beginning of diseases, tracking the development of chronic ailments, and pinpointing risk factors for unfavorable health outcomes. The following ways demonstrate the use of predictive analytics in IoMT (Mohanty and Pescador 2021):

- **Early disease detection**: To find patterns and trends associated with certain diseases or cases, predictive analytics algorithms may be trained on enormous datasets of patient health records and medical sensor data. Even before the patient exhibits any symptoms, they can examine this data to assess their likelihood of getting a specific disease (Manickam et al. 2022). For instance, an electrocardiogram (ECG)–based predictive analytics algorithm could spot patterns in a patient's heart rhythms that suggest a patient is more likely to experience a heart problem. This would enable medical professionals to respond quickly and possibly delay the beginning of the ailment or lessen its severity

(Ahmed et al. 2021).

- **Monitoring the advancement of diseases**: Predictive analytics algorithms can be used to keep track of long-term disorders' progression, like diabetes, hypertension, or asthma. By analyzing patient health data obtained from wearables and medical sensors, they can forecast the course of these disorders and identify prospective health hazards that might pose problems (Manickam et al. 2022). For instance, a predictive analytics model trained on blood glucose data might spot trends in glucose levels that suggest a patient may have hypoglycemia or hyperglycemia. By adjusting the patient's treatment plan as a result, medical experts may be able to avoid significant problems (Cao et al. 2020).
- **Treatment optimization**: By examining a patient's health data and medical records, predictive analytics algorithms can identify the most effective therapies for particular ailments and estimate the outcomes of various treatment choices. For instance, a patient with hypertension might receive a personalized treatment plan from a predictive analytics model that has been trained on patient health data and medical records. The model could forecast the effectiveness of various medications, doses, and lifestyle modifications, enabling medical professionals to design a treatment strategy tailored to each patient's needs (Amin et al. 2019).
- **Risk stratification**: Predictive analytics algorithms can detect risk factors and forecast the likelihood of a patient having a specific ailment by examining patient health data and medical records (Khamparia et al. 2021). A predictive analytics model, for instance, may be able to pinpoint patients who are at a high risk of developing heart disease after being trained on patient health data and medical records. The approach might assist healthcare professionals in developing individualized preventative plans for their patients by analyzing risk factors, including age, weight, blood pressure, cholesterol levels, and family history (Shanmugam and Azam 2023; Zhou et al. 2020).

In the context of the IoMT, several predictive analytics algorithms can be utilized to forecast future health outcomes, pinpoint potential health hazards, and enhance treatment regimens for specific patients. The following are some examples of predictive analytics algorithms frequently used in IoMT:

- **Logistic Regression (LR)**: It is a statistical procedure used to estimate the likelihood of a binary outcome. LR can be applied in IoMT to identify and predict risk factors that increase a patient's likelihood of developing a specific illness (Nagarajan et al. 2021). It has been employed to estimate a patient's risk of developing heart disease based on their medical history and risk variables such as age, gender, blood pressure, cholesterol levels, and genetics (Khamparia et al. 2021). In (Qian 2019), the LR model was developed using data from nearly 3000 patients to predict the risk of heart disease with up to 85% accuracy. A diagnostic system for heart disease was also implemented in Divya et al. (2021), which utilized logistic regression among other algorithms within the IoMT framework to predict heart disease risk. This system employed the Cleveland cardiac dataset from the UCI research repository, achieving an accuracy of 88.59% with logistic regression using a majority voting approach.
- **Random Forest (RF)**: A different random subset of the input data is used to construct each decision tree's functional structure. RF can be utilized in the Internet of Medical Things (IoMT) to forecast the course of chronic illnesses like diabetes and hypertension

based on patients' health information and medical records. In Qian (2019), researchers utilized data from more than 1,000 diabetic patients to train an RF model that achieved an accuracy of 75% in predicting changes in blood glucose levels (Khamparia et al. 2021). To ensure accurate illness detection, (Fawagreh and Gaber 2020) created a forecasting model for healthcare data analytics using a pruned RF technique. The goal is to increase the RF algorithm's computational effectiveness and prediction speed while preserving its excellent predictive accuracy, especially for healthcare applications. Ooka et al. (2021) aimed to predict changes in glycated hemoglobin A1c (HbA1c) levels using an RF technique for early type 2 diabetes prediction. The trial included 42,908 participants with HbA1c levels below 6.5% who were not undergoing diabetic therapy. In Elbasi and Zreikat (2021), patient data were gathered from various IoMT devices, including laboratories, medical imaging, doctor reports, ambulances, and wearable technology. These data were then categorized and analyzed for diagnosis and treatment using machine learning algorithms. The findings showed that the RF approach yielded more than 93% accuracy.

- **Neural Networks (NNs)**: Based on information from medical sensors and patient health records, NNs in IoMT can anticipate the beginning of particular medical disorders. Yellamma et al. (2020) presented a neural network method that uses 10 characteristics from a WBC dataset of 699 patients as input to predict the presence of breast cancer. Using GD, the method attempted to minimize the error in a particular weight space. This model's accuracy was as high as 98.9%. In Prabakaran et al. (2021), a neural network model was built using data from over 800 patients that was 90% accurate in predicting the beginning of Parkinson's disease (Sinha et al. 2023). Implementing a support value-based deep neural network (SDNN) for the precise classification and early identification of malignant cells in brain images in e-Health care was described in Vankdothu et al. (2022). According to experimental data, the suggested SDNN technique achieved an accuracy of up to 94.30%.
- **Support Vector Machine (SVM)**: Based on medical sensor data, medical records, and other pertinent criteria, SVM has been employed in IoMT to forecast the likelihood that individuals would experience the onset of various health disorders. In Tang (2020), scientists used information from more than 2,000 patients to train a SVM model that was able to predict the likelihood of getting hypertension with an accuracy of 85% (Sinha et al. 2023; Abdelaziz et al. 2019). Thakur et al. (2023) focused on applying ensemble ML models, which incorporate various ML techniques, to improve diabetes prediction during the COVID-19 pandemic. The study used the Pima Indians Diabetes Database. The accuracy of the SVM classifier was a promising 76.62%. Methods for generating polynomial and interaction features were employed, and the results showed strong correlations, especially with characteristics like glucose. The ensemble model, which included Gradient Boosting, SVM, and Decision Trees, produced an accuracy of 93.2%, which is a significant gain above single classifiers.

Predictive analytics in the IoMT is making significant strides in enhancing healthcare outcomes by leveraging ML and DL methodologies. Table 2 reviews recent studies that have applied predictive analytics to IoMT, showcasing a variety of techniques and their applications in healthcare. The studies spanned a range of methodologies, including distributed hierarchical DL, logistic regression, RF, SVM, and neural networks. These methodologies are applied to diverse datasets, such as patient health records, biosignals from wearable devices, and medical imaging data, to predict various health conditions like heart disease, diabetes, cancer, and hypertension. Integrating these advanced analytical techniques with IoMT devices facilitates real-time monitoring, early disease detection, and personalized treatment plans, thereby improving patient management and preventive care. However, challenges such as data privacy, computational resource requirements, and model interpretability highlight the need for further research and development to optimize these predictive models for broader applicability and integration into existing healthcare systems.

2.2 Natural language processing (NLP)

NLP, which is the focus of the AI field, is the processing and analysis of human language. In the context of the IoMT, NLP algorithms can be used to evaluate and comprehend unstructured text data from various sources, including medical records, clinical notes, and patient-generated data. Certain NLP's applications in IoMT can be clarified in the following: (Qian 2019; Prabakaran et al. 2021):

- Extracting crucial data from clinical documentation about patients, including their history, diagnosis, treatments, and procedures. This data then be used for research, quality control, and clinical decision-making (Nagarajan et al. 2021).
- Analyzing patient-generated data, such as patients' comments, social media postings, and online forum posts. In order to develop patient-centered care strategies, this data can be utilized to discover the patient's requirements, preferences, and concerns.
- Developing clinical decision support tools that provide clinicians with pertinent information at the time of care. For instance, clinical notes can be examined using NLP, and recommendations can be made for diagnostic procedures, medications, and other therapies in light of the patient's medical history and present symptoms.
- Evaluating unstructured text data and finding patterns and trends that can be utilized to guide the development of predictive analytics models. NLP, for instance, can be used to analyze social media data to spot trends in public health, such as infectious disease epidemics.

A variety of NLP algorithms are applicable to the IoMT. The following are the most popular NLP algorithms that can be applied in IoMT (Lakshmanaprabu et al. 2019):

- **Named Entity Recognition (NER)**: It is an algorithm for locating and extracting named entities from unstructured text data, including names of people, locations, and medical terminology (Zhou et al. 2020; Prabakaran et al. 2021). NER can be used in IoMT to extract vital data from clinical notes and patient-generated data, like patient history, diagnosis, drugs, and treatments. In (Qian 2019), NER was employed by researchers to accurately extract crucial medical ideas from EHRs and medical concepts from unstructured clinical notes, including symptoms, diagnoses, and prescriptions. This led to a rise in clinical decision-making accuracy (Hassan et al. 2021).
- **Sentiment analysis**: It is an algorithm for determining whether a piece of text includes a good, negative, or neutral sentiment (Qian 2019). Sentiment analysis in IoMT can be used to examine patient comments, social media posts, and online forums to learn about

Table 2 Recent studies on predictive analytics with IoMT

Refs.	Year	Methodology	Datasets Used	Advantages	Limitations
Qian (2019)	2019	A distributed hierarchical DL system over a cloud server and smartphone for predicting risk of heart disease based on medical history and risk factors	data from nearly 3000 patients to predict the risk of heart disease with up to 85% accuracy	Helps identify patients at risk and improve preventive care. Enables timely intervention through real-time fall monitoring	High complexity in developing and managing the distributed DL system. Potential concerns related to the handling and security of personal and sensitive data
Divya et al. (2021)	2021	A diagnosis system utilizing logistic regression to foresee heart disease risk	The UCI research repository's Cleveland cardiac dataset includes patient health records, with specific metrics such as heart rate, blood pressure, and ECG readings, collected via IoMT devices	High diagnostic accuracy (88.59%). Real-time monitoring and diagnosis. Enhanced patient management	Data privacy concerns. Variability in data quality from different sources
Qian (2019)	2019	A distributed hierarchical DL system for predicting changes in blood glucose levels in patients with diabetes	Data from more than 1000 diabetic patients to train an RF model that had an accuracy of 75% for predicting changes in blood glucose levels	Helps optimize treatment plans and reduce the risk of complications. Enhances scalability and efficiency by leveraging multiple computing nodes	Limited computational resources on wearable devices may affect DL performance. The system may be optimized for specific types of falls or user populations, affecting broader applicability
Fawagreh and Gaber (2020)	2020	Forecast model for healthcare data analytics using a pruned RF technique to ensure accurate illness detection	Three medical regression data sets of three different diseases. These diseases are Parkinson's, diabetes, and breast cancer. The first data set was obtained from the University of Waikato Repository, and the last two from the University of California, Irvine (UCI) ML Repository	Increase the RF algorithm's computational effectiveness and prediction speed while preserving its excellent predictive accuracy, especially for healthcare applications	The effectiveness of clustering depends on the chosen method. Results may vary across different healthcare datasets. The process of pruning and clustering adds complexity to the model compared to traditional RF

Table 2 (continued)

Refs.	Year	Methodology	Datasets Used	Advantages	Limitations
Ooka et al. (2021)	2021	A RF technique to predict changes in glycated hemoglobin A1c (HbA1c) levels for early type 2 diabetes prediction	The study comprised 42,908 participants with HbA1c levels less than 6.5% who were not undergoing diabetic therapy. The dataset consists of 51 health-check items along with their changes from the previous year. The primary outcome is the change in HbA1c levels over the next year	RF demonstrated superior predictive accuracy for changes in HbA1c. It can manage a large number of predictors. Provides insights into the relative importance of different predictors. It can include various variables and handle nonlinear relationships between predictors and outcomes	RF can be less interpretable than simpler methods like MLR. The results may be specific to the dataset and population used in the study (Japanese health check-up data). RF models can be computationally intensive, which might be a concern for resource-constrained environments
Elbasi and Zreikat (2021)	2021	Enhance early prediction and diagnosis of diseases using various ML algorithms (e.g., classification, regression) applied to IoMT data	Patient data collected via IoMT devices, including laboratories, medical imaging, reports from doctors, ambulances, and wearable technology	Early detection of diseases, leading to better health outcomes. ML algorithms can analyze large volumes of IoMT data efficiently, improving diagnostic accuracy and speed. Methods can be adapted to various IoMT data types and different healthcare scenarios	Integrating ML algorithms with existing IoMT systems may pose technical and practical challenges. Analyzing sensitive health data raises privacy and security concerns. Models trained on specific datasets may not generalize well to other datasets or real-world scenarios without further validation
Yellamma et al. (2020)	2020	A diagnostic neural network model to predict the presence of breast cancer using Multi-Layer Perceptron (MLP) with backpropagation	WBC breast cancer data likely includes features such as tumor size, texture, and other relevant characteristics	An automated approach could potentially improve early breast cancer detection. MLP with backpropagation can achieve high accuracy by learning complex patterns from data. Backpropagation allows the model to adjust weights and biases effectively during training	MLP effectiveness model depends on the quality of the training dataset. MLPs may be less interpretable than traditional methods, making it harder to understand the decision-making process. Training and optimizing MLP can be resource-intensive, requiring significant computational power and time

Table 2 (continued)

Refs.	Year	Methodology	Datasets Used	Advantages	Limitations
Prabakaran et al. (2021)	2021	Deep NNs model for predicting the beginning of Parkinson's disease and processing biosignals from wearable devices. Introduces a BioNetExplorer tool designed to explore various neural network architectures and optimize them for biosignal processing	Biosignal data collected from wearable devices, which may include signals such as ECG, EEG, or accelerometer data. Specific datasets are not detailed in the abstract	Optimizing neural network architectures specifically for biosignal processing, leading to more efficient and accurate models. Improving the performance of biosignal processing tasks. The tool can be applied to various types of biosignals and different wearable applications	Optimizing neural network architectures can be complex and computationally intensive. Models optimized for specific biosignal types may not generalize well to other types without adjustments. Integrating optimized architectures into actual wearable devices may face hardware challenges
Vankdothu et al. (2022)	2022	A support value-based deep neural network (SDNN) for the precise classification and early identification of malignant cells in brain images in e-Health care	Dataset consists of brain images, including MRI, CT scans, or other neuroimaging modalities	The support value-based approach improves the accuracy and reliability of brain image classification. Leveraging IoMT allows for efficient and scalable processing of medical images. Utilizes advanced DL techniques to handle complex image data and improve diagnostic capabilities	The model's performance depends on the quality of the brain image dataset. Training deep NNs requires significant computational resources and may involve long processing times. Deep NNs can be complex and less interpretable. The model's effectiveness may vary with different brain image types
Tang (2020)	2020	A smart logistics model leveraging IoT technology to predict the risk of developing hypertension based on medical sensor data and medical records	Data related to logistics operations	Enables early intervention and prevention of hypertension. Reducing healthcare costs	Integrating IoT into existing logistics systems can be complex and require significant investment. Managing vast amounts of data presents challenges related to privacy and cybersecurity
Chidambaranathan et al. (2021)	2021	SVM model for classifying brain tumor MRI images within a Cloud IoMT framework	Dataset consists of MRI images of brain tumors, including various types and stages of tumors	Optimized SVM provides high accuracy in classifying brain tumor images, improving diagnostic capabilities. Cloud integration allows for scalable processing and storage. Cloud-based deployment ensures easy access to the model and data from various locations and devices	Cloud-based solutions require reliable internet and substantial computational resources, which can be costly. Handling sensitive medical data in the cloud concerns data privacy and security. Optimizing SVM models and integrating them into the cloud can be complex

Table 2 (continued)

Refs.	Year	Methodology	Datasets Used	Advantages	Limitations
Thakur et al. (2023)	2023	An enhanced model for predicting diabetes amidst the COVID-19 pandemic using ensemble ML techniques	Pima Indians Diabetes Database includes health records with features relevant to diabetes prediction, possibly augmented with data related to COVID-19 impacts	Ensemble models provide better prediction performance and enhance the robustness and reliability of the prediction model compared to individual models. Addresses the added complexity of predicting diabetes during the COVID-19 pandemic	Model performance depends on the quality of the used health data. Ensemble models can be more complex and computationally intensive compared to single models. Integrating the model into real-world healthcare systems and ensuring its usability by healthcare professionals can be challenging

Table 3 Recent studies on NLP algorithms with IoMT

Refs.	Year	Methodology	Datasets Used	Advantages	Limitations
Qian (2019)	2019	Extracting crucial medical concepts from EHRs and patient-generated data employing NER	EHRs and medical concepts from unstructured clinical notes, including symptoms, diagnoses, and prescriptions	Improved accuracy in clinical decision-making	Potential concerns related to the handling and securing personal and sensitive data. The system may be optimized for specific types of falls or user populations, affecting broader applicability
Gligic et al. (2020)	2020	Transfer learning model with bootstrapped neural networks for NER in EHRs	Clinical Notes, and EHR Systems	Leveraging pre-trained models enhances accuracy. Transfer learning reduces the time needed. The combination of transfer learning and bootstrapping allows the model to better adapt to specific tasks and datasets	There is a potential for overfitting to the bootstrapped training subsets. Requires careful tuning and integration of both transfer learning and bootstrapping techniques. EHRs can be highly variable in format and content, which may complicate model training and generalization
Pearson et al. (2021)	2021	A combination of NLP techniques and ML algorithms (e.g., CRF, BiLSTM, or transformer-based models like BERT) for NER in unstructured medical text	Clinical notes, medical records, and publicly available datasets like MIMIC-III and i2b2	Extract relevant medical information from unstructured text. Reduces the manual effort required for data annotation and information extraction. Advanced models, especially those using DL, tend to achieve higher accuracy in recognizing named entities	Requires significant computational power for training complex models. DL models, while accurate, can be difficult to interpret and understand
Yang et al. (2022)	2022	A deep neural network for NER, involving data preprocessing, model architecture (e.g., LSTM or BERT), training, and evaluation of medical text data	Medical text datasets such as clinical notes, medical records, or specialized datasets like MIMIC-III or i2b2	DL models often outperform traditional approaches in NER tasks. Processing large volumes of text data efficiently. Automation of annotation processes, saving time and resources	Performance heavily relies on the quality of training data. Requires significant computational resources and expertise. Models trained on specific datasets may not perform well on different datasets
Prabakaran et al. (2021)	2021	Sentiment analysis method to examine patient reviews of mobile health applications	Patient feedback, social media posts, and online forums	Patients who submitted favorable reviews were more likely to mention improved health outcomes and higher adherence to treatment plans	

Table 3 (continued)

Refs.	Year	Methodology	Datasets Used	Advantages	Limitations
Pathuri et al. (2021)	2021	Feature-based sentiment analysis approach with the CUDA-SADBM classification model. Integrated CUDA (Compute Unified Device Architecture) for accelerated computing to enhance the performance of the SADBM (Self-Adaptive Deep Boltzmann Machine) model in analyzing public attention towards COVID-19	Data related to public attention on COVID-19, including social media posts, news articles, and public surveys	Utilized feature-based methods for detailed sentiment analysis. CUDA acceleration improves computational efficiency, processing speed, and model performance	Integration of CUDA and SADBM introduces complexity in implementation and tuning. High computational demands for training and running the model. Dependence on the quality and representativeness of the dataset
Jalil et al. (2022)	2022	ML and DL techniques, such as transformers, neural networks, and ensemble methods, for sentiment analysis on COVID-19-related texts	Data related to COVID-19, including social media posts, news articles, and public opinions	Utilized advanced techniques for high accuracy and nuanced sentiment detection. Effective in handling diverse and large-scale COVID-19-related texts	Potential challenges with data quality and consistency. High computational requirements for training complex models
Xu et al. (2023)	2023	Adversarial learning to enhance sentiment analysis for IoMT systems, focusing on improving the robustness of sentiment analysis models against perturbations and adversarial attacks	Datasets related to IoMT systems, including social media data, user feedback, and discussions from medical forums related to IoMT systems	Improves the model's ability to handle noisy and adversarial inputs. Better sentiment analysis performance due to increased model resilience. Tailored techniques for analyzing sentiments in the context of IoMT systems	Complexity in implementation and tuning. Dependence on data quality, diversity of the training data, and computational resources
Tang (2020)	2020	Individualized medication instructions for patients with inadequate health literacy using NLG	Patient-facing communications, such as discharge summaries, medication instructions, and patient education materials	Patients who received tailored instructions followed their treatment plans, which enhanced their health results. Improving patient engagement and adherence to treatment plans	Managing and securing sensitive data can pose challenges. Initial setup and maintenance of systems may involve significant investment
Hoogi et al. (2020)	2020	NLG model for generating simulated mammography reports using NLG techniques trained on existing report data	dataset consisted of real mammography reports used to train the NLG model, including report texts and clinical data	Generates realistic mammography reports for research without using sensitive patient data. Provides a means to test and refine systems that rely on medical report generation. It is useful for professionals and ML models on mammography report formats	The quality of generated reports may vary and may not fully capture all nuances of real-world clinical reports. The model's effectiveness is limited by the quality of the training dataset

Table 3 (continued)

Refs.	Year	Methodology	Datasets Used	Advantages	Limitations
Zhou et al. (2020)	2020	Word embedding to analyze medical text data and identify semantic relationships between medical terms	Data from IoHT devices, including wearable sensors and motion data	Word embedding could precisely detect the semantic connections between medical phrases, improving the accuracy of clinical judgment	Requires significant computational resources for DL models. Dependent on the quality and variety of sensor data
Koutsomitropoulos and Andriopoulos (2020)	2020	An automated system to automate the indexing of biomedical literature using contextualized word representations. Employed advanced word embedding techniques to improve indexing accuracy using the Medical Subject Headings (MeSH) system	Biomedical literature dataset included texts from various biomedical sources, including research articles and medical papers	Enhance the precision of MeSH term assignment by considering the context in which terms are used. Automation reduces the manual effort for indexing large biomedical literature. Provides a standardized approach to indexing, ensuring consistent application of MeSH terms	Effectiveness depends on the quality and coverage of the training data for word embeddings. May require substantial computational resources for processing large volumes of text
Habib et al. (2021)	2021	A specialized word embedding model, called AltibbiVec, specifically designed for medical and health applications in the Arabic language. The model leverages word embeddings to transform Arabic medical and health-related terms into numerical vectors that capture semantic relationships and contextual meanings	Arabic medical and health texts, including Arabic-language medical literature, clinical records, and health-related documents	Improves the performance of NLP tasks related to medical texts in Arabic. Provides better semantic understanding of medical terms and concepts within the Arabic language. Tailored specifically for medical and health contexts, improving accuracy and relevance in these applications	The effectiveness depends on the quality and size of the Arabic medical dataset used. The model is specialized for Arabic and may not be applicable to other languages. May require adaptation for other languages or other specialized health domains or non-medical contexts

the needs, preferences, and worries of patients (Mishra and Tyagi 2022). In Prabakaran et al. (2021), researchers used sentiment analysis to examine patient reviews of mobile health applications. The researchers discovered that patients who submitted favorable reviews were likelier to mention improved health outcomes and higher adherence to treatment plans (Wang et al. 2020).

- **Natural Language Generation (NLG)**: It is an algorithm for generating text that mimics human speech based on rules and data. NLG can be utilized in IoMT to produce patient-aimed messages, including summaries of discharges, directions for taking medications, and patient education materials (Zhou et al. 2020; Hassan et al. 2021). In Tang (2020), researchers provided individualized medication instructions for patients with inadequate health literacy using NLG. They discovered that patients who received tailored prescription instructions followed their treatment plans more frequently, which enhanced their health results (Wang et al. 2020).
- **Word embedding**: It is an algorithm for representing words as vectors in a high-dimensional space. To increase the precision of clinical decision-making, word embedding can be utilized in IoMT to evaluate medical text data and uncover semantic links between medical terms (Tang 2020). In Zhou et al. (2020), researchers utilized word embedding to examine medical text data and discover semantic connections between medical terms. They discovered that word embedding could precisely detect the semantic connections between medical phrases, improving the accuracy of clinical judgment.

The advent of the IoMT has opened up new avenues for applying NLP algorithms to enhance healthcare delivery and patient outcomes. Table 3 reviews recent studies focusing on the intersection of NLP and IoMT, highlighting various methodologies, datasets, advantages, and limitations. These studies encompassed various applications, including NER from EHRs, sentiment analysis of patient feedback and public opinion, and generating natural language for medical reports and patient instructions. By leveraging advanced techniques such as transfer learning, DL, adversarial learning, and word embedding, these studies aimed to improve clinical decision-making, patient engagement, and the automation of medical text processing. Integrating NLP with IoMT not only enhances the precision and efficiency of healthcare systems but also addresses challenges related to data quality, computational resources, and model robustness in diverse clinical and linguistic contexts.

2.3 Image and signal processing

IoMT relies heavily on image and signal processing to record, analyze, and interpret medical images and data in real-time. Using cutting-edge algorithms and methodologies, image and signal processing algorithms can help improve the precision of medical diagnoses, monitor patient health, and enable new types of remote patient care (Raj et al. 2020). Images from X-rays, MRI scans, and ultrasounds can be analyzed and understood using image-processing algorithms in the context of IoMT. For instance, computer-aided detection algorithms can detect early cancer indications in medical images, leading to an earlier diagnosis and better patient outcomes. Similarly, by training ML algorithms on big datasets of medical images, they can be enhanced in their ability to recognize and diagnose a variety of medical disorders (Huang et al. 2021).

On the other hand, physiological signals like electrocardiograms (ECGs), electroencephalograms (EEGs), and electromyograms (EMGs) can be analyzed and interpreted using signal processing algorithms (Sutradhar et al. 2023; Devi et al. 2023). For instance, heart rate variability in people with heart disease can be tracked based on signal processing algorithms, giving early indications of potential cardiac events (Huang et al. 2021). Massive datasets of physiological signals may also be used to train ML algorithms, which will improve their accuracy in recognizing and diagnosing a variety of medical disorders (Li et al. 2021; Betty Jane and Ganesh 2020).

One of the major advantages of image and signal processing in IoMT is real-time patient health monitoring, which enables early disease detection and treatment. For instance, wearable technology that records physiological signals, like ECGs and EMGs, could track patients with long-term illnesses like diabetes and hypertension, providing them with real-time feedback and alarms for potential health problems. Additionally, real-time analysis of medical images using image-processing algorithms enables medical personnel to make quick, well-informed judgments about patient care (Raj et al. 2020). There are a variety of algorithms used in image and signal processing for IoMT, as follows:

- **Image segmentation**: It is an algorithm for dividing an image into several regions or segments that can be used to recognize objects or other interesting areas in an image. Image segmentation is frequently used in medical imaging to spot lesions or malignancies (Ali et al. 2023). In Raj et al. (2020), brain cancers were accurately identified from MRI scans with an accuracy of 85% using image segmentation algorithms, which improved treatment outcomes by enabling early brain tumor detection and diagnosis (Wang et al. 2020; Ali et al. 2023).
- **Object recognition**: Objects in the image, such as tumors, blood arteries, or other anatomical structures, can be recognized with the aid of object recognition algorithms (Christal et al. 2023). Through training these algorithms on enormous databases of medical images, their accuracy can be enhanced in identifying and diagnosing medical diseases. With an accuracy of 85%, researchers from the University of California, San Francisco, used object recognition algorithms to classify and identify lung nodules from CT scans (Huang et al. 2021), which increased survival rates by enabling early lung cancer detection and diagnosis (Christal et al. 2023).
- **Classification**: Medical images, based on their properties, can be categorized into groups, such as healthy versus diseased or benign versus malignant, using classification algorithms. These algorithms can be applied to medical imaging for diagnosing diseases like Alzheimer's disease and cancer (Cao et al. 2020; Datta Gupta et al. 2023). With 94% accuracy, researchers from the University of Oxford have been able to classify retinal images and identify diabetic retinopathy (Cao et al. 2020), which enhanced patient outcomes by enabling medical professionals to recognize and diagnose diabetic retinopathy early (Datta Gupta et al. 2023).
- **Feature extraction**: It is an algorithm for extracting certain features or patterns from an image. After that, utilizing these extracted features, the classification or medical disease diagnosis can be made. Using feature extraction algorithms, researchers from Stanford University could pinpoint and track particular aspects of skin lesions from photos captured by a smartphone camera (Raj et al. 2020). Clinicians have been able to detect skin cancer early because of the ability to trace the evolution of skin lesions over time.

- **Signal processing**: Physiological signals can be analyzed relying on signal-processing algorithms that can be used to track variations in heart rate, spot irregularities in brain activity, or monitor muscular activity (Cao et al. 2020). In Huang et al. (2021), signal processing algorithms were utilized to examine the ECG signals from patients with heart failure, which improved patient outcomes since clinicians could track heart rate variability and spot changes in heart function over time.Integrating IoMT with advanced image and signal processing techniques has significantly transformed the healthcare industry. Recent studies have delved into various methodologies to enhance medical image analysis's accuracy, efficiency, and reliability. Table 4 presents an overview of notable research contributions in this field, detailing their methodologies, datasets used, advantages, and limitations. The studies encompassed a range of applications, including tumor detection, medical image segmentation, object recognition, and disease classification, showcasing the diverse potential of IoMT in improving diagnostic and treatment outcomes. These advancements highlighted the crucial role of computational resources, dataset quality, and model robustness in achieving real-time, precise medical imaging solutions within the IoMT framework. By leveraging cutting-edge technologies such as DL, transfer learning, and augmented reality, these studies demonstrate the significant strides made towards more effective and efficient healthcare delivery.

2.4 CNN

IoMT technologies, including the network of wearables, connected medical devices, remote monitoring systems, and other digital technologies, collect and produce enormous volumes of health data that may be examined to provide information about the health and behavior of patients and to support clinical decision-making (Huang et al. 2021). DL algorithms can be used to assess these huge and complicated volumes of health data produced by IoMT technologies (Al-Sa'D et al. 2018; Khan et al. 2023). These DL algorithms can be trained to recognize patterns, anticipate outcomes, and pinpoint abnormalities. For instance, DL can be utilized to analyze wearable device data about vital signs to spot early warning signals of illness or irregularities that may warrant medical attention. As a way to support professional judgment and enhance patient outcomes, DL can also be used for examining electronic health information or medical imaging (Huang et al. 2021).

One of the most widespread DL algorithms for image and video processing is the CNN model (Lakshmanaprabu et al. 2019). CNN model is designed to automatically and effectively learn spatial hierarchies of features from the input data in order to recognize patterns and objects in images properly. The three main components of a CNN are convolutional layers, pooling layers, and fully linked layers. The input image properties, including edges, forms, and textures, are extracted using filters applied by convolutional layers to the input image (Al-Sa'D et al. 2018). Pooling layers are used to downsample the output of the convolutional layer, which shrinks the spatial size of the feature maps while preserving the most crucial data. Fully linked layers employ the output from convolutional and pooling layers to perform classification or regression operations. In various image recognition and computer vision applications, including object detection, picture classification, and semantic segmentation, CNNs have produced outstanding results (Raj et al. 2020).

VGG-16 and AlexNet are well-liked CNN models, which are frequently used in computer vision applications like semantic segmentation, object identification, and picture clas-

Table 4 Recent studies on image and signal processing with IoMT

Refs.	Year	Methodology	Datasets Used	Advantages	Limitations
Raj et al. (2020)	2020	Image segmentation algorithm for medical image classification and identifying tumors from MRI scans using optimal feature selection mechanism to identify the most relevant features from medical images	Various medical images, including X-rays, MRI scans, CT scans, and ultrasound images	Improves treatment outcomes by enabling early brain tumor detection and diagnosis. Improves classification accuracy through optimal feature selection. Enhanced processing efficiency by focusing on relevant features. Facilitates real-time image analysis and decision-making	Requires significant computational resources for training. Performance is dependent on the quality and diversity of the dataset. Needs further validation for generalization across different medical images and settings
Tang et al. (2022)	2022	A unified framework for medical image segmentation that learns from uncertainty in an end-to-end manner, incorporating uncertainty estimation directly into the segmentation process	Diverse types of medical images, likely including MRI scans, CT scans, ultrasound images, and X-rays	Enhances segmentation accuracy by handling ambiguous regions effectively. The model's ability to learn from uncertainty makes it more robust to variations in medical images. Adaptable to different types of medical images and segmentation tasks	The inclusion of uncertainty estimation adds computational complexity. Requires extensive validation across different clinical settings
Ali et al. (2023)	2023	An IoMT-based system for precise melanoma lesion segmentation from medical images using conditional generative adversarial networks (cGANs)	Medical images of skin lesions, likely including labeled datasets of melanoma and non-melanoma images for training and validation	Improves accuracy in segmenting melanoma lesions. Utilizes cGANs to improve the precision of lesion boundaries, which is crucial for accurate diagnosis. Real-time segmentation within the IoMT framework for timely diagnosis and treatment	Requires substantial computational resources for training cGANs. Performance depends on the quality of the training dataset. Needs validation across diverse clinical settings to ensure robustness
Huang et al. (2021)	2021	Object recognition algorithms to classify and identify lung nodules from CT scans	Dataset of medical materials, demographic data, characteristics of past emergencies, and supply chain logistics information	Increases survival rates by enabling early lung cancer detection and diagnosis	Relies on quality historical data. Requires significant computational resources. Needs frequent updates for new emergencies
Jin et al. (2022)	2022	Anatomy-guided DL framework for object recognition in medical images	A large collection of annotated medical images from various sources, such as MRI, CT, and X-rays, covering different anatomical regions and medical conditions. Detailed annotations of anatomical structures and objects of interest	Incorporating anatomical guidance improves object recognition accuracy. The model is more robust to variations in medical images. Better alignment with clinical knowledge and practices	Performance relies on the quality of the annotated data. The integration of anatomical information increases the complexity. The model may need adjustments to generalize across different imaging modalities

Table 4 (continued)

Refs.	Year	Methodology	Datasets Used	Advantages	Limitations
Rupa et al. (2022)	2022	A method for detecting medicine drug names from images of medicine packages using object recognition techniques integrated with augmented reality (AR)	Images of medicine packages, which are collected from various sources, containing different brands and types of medicine. Annotations, including manual labeling of drug names on the medicine packages, to train the recognition model	The AR application aids in the easy and quick identification of medicines. The integration with AR allows for real-time detection and overlay of drug names. The application is user-friendly, making it accessible to non-expert users	Performance is affected by lighting conditions and image angles. Real-time processing needs significant resources, limiting use on low-end devices. It requires updates on new drug names and packaging designs
Fengping et al. (2021)	2021	A medical image classification algorithm that incorporates a visual attention mechanism to enhance the model's focus on relevant parts of the medical images within a multi-channel CNN (MCNN) to process different types of features and channels in medical images	Dataset comprises medical images from various sources, including different modalities such as MRI, CT scans, or X-rays	The visual attention mechanism helps the model focus on critical image areas, leading to better classification performance. MCNN architecture allows for effective extraction and utilization of multiple feature types, improving overall model robustness	Requires significant resources for training and inference. May need adaptation for new types of images or conditions
Mabrouk et al. (2022)	2022	A pre-trained DL model, adapting for medical image classification. a transfer learning (TL)-based method for feature extraction was employed, which is carried out using MobileNetV3. Chaos Game Optimization (CGO) techniques were applied to optimize the hyperparameters and architecture of the model	A variety of medical images, potentially sourced from different imaging modalities like X-rays, MRIs, and CT scans, with annotated labels	The model is designed to work within the IoMT framework, facilitating real-time medical image classification. Transfer learning combined with CGO improves classification accuracy. CGO optimizes model performance, aiding real-time applications	Relies on the quality and diversity of the dataset. The optimization process is resource-intensive. May need further fine-tuning for specific applications or new conditions
Ren et al. (2023)	2023	DL network based on ResNet-50 merged transformer named RMT-Net for classifying COVID-19 medical images. The Transformer model leverages self-attention mechanisms to capture global dependencies in the image data	COVID-19 medical images, which include various types of medical images related to COVID-19, such as chest X-rays and CT scans	Transformer-based model improves classification accuracy by capturing long-range dependencies and complex patterns in medical images. The model can be adapted to different types of medical imaging tasks beyond COVID-19	Transformers can be computationally intensive, requiring substantial resources for training and inference. The model may need adjustments to perform well on medical images outside the COVID-19 domain or in different imaging conditions

Table 4 (continued)

Refs.	Year	Methodology	Datasets Used	Advantages	Limitations
Song et al. (2021)	2021	Feature extraction method using a neural network algorithm to process and fuse features from different types of medical images to produce enhanced, comprehensive images, which has high feature extraction effect on medical images of chest, lung, brain, and liver	Various medical images, such as MRI and CT scans, are sourced from publicly available medical imaging databases or collected from medical institutions	The fusion process improves the overall image quality by combining features from different sources. Provides comprehensive images that highlight important details, aiding in more accurate diagnoses. The use of a neural network automates the feature extraction and fusion process	Neural network algorithms can be complex and require significant computational resources. The effectiveness depends on the quality and diversity of the training data. Requires further validation on diverse datasets to ensure broad applicability
Kuwil (2022)	2022	Feature extraction approach for medical images based on data distribution skew, called Feature Extraction Based on Region of Mines (FE_mines). The methodology involves analyzing the asymmetry in pixel intensity distributions within the images	Various medical images, such as MRI, CT scans, and X-rays, are sourced from publicly available databases or medical institutions. The study likely includes a significant number of images, covering both normal and pathological cases, with standard pre-processing applied	The method can effectively highlight important features in medical images, improving diagnostic accuracy. It is robust to variations in imaging conditions and noise, making it versatile across different imaging conditions. It is computationally efficient and suitable for real-time clinical applications	May be optimized for specific types of medical images and less effective on others. The effectiveness of the approach depends on the quality and consistency of the input images. It requires further validation on diverse datasets and in clinical settings to confirm its broader applicability
Aileni et al. (2020)	2020	Daubechies discrete wavelet transform (DWT) with correlation analysis of biomedical signals, such as electroencephalogram (EEG) measurements of brain electrical activity, electromyogram (EMG) measurements of muscle electrical currents, and photoplethysmography (PPG) measurements of heart rate. ANN was employed to automatically identify epileptic episodes before their occurrence	EEG signals recorded from various subjects and collected from EEG monitoring systems or public EEG datasets. 30 EEG recordings 10 min were analyzed before a seizure and during the seizure for 30 patients with epilepsy	Provides detailed insights into brain activity through advanced ANN analysis. Enable accurate prediction of brain activity patterns. Automates the analysis process, reducing manual effort and potential errors	The effectiveness of the ANN models depends on the quality and accuracy of the EEG data. Training and applying ANNs can be computationally intensive. The models may need further validation across different datasets and conditions to ensure robustness

Table 4 (continued)

Refs.	Year	Methodology	Datasets Used	Advantages	Limitations
Zhang et al. (2023)	2023	Method to infer heart rate variability (HRV) by integrating signal processing techniques with ML algorithms. Signal processing pre-processes the raw heart rate data to remove noise and extract relevant features. ML models are applied to analyze the pre-processed data and directly infer HRV metrics	A large dataset of photoplethysmography (PPG) signals HRV ground truth. Heart rate data from wearable devices or clinical equipment	Combines signal processing with ML for rapid and accurate HRV inference. Improves the accuracy of HRV measurement. Automates the HRV analysis process, reducing the need for manual intervention	The method's accuracy depends on the quality of the input heart rate data. Requires expertise in both signal processing and ML for implementation

Table 5 Recent studies on CNN models with IoMT

Refs.	Year	Methodology	Datasets Used	Advantages	Limitations
Demirel et al. (2021)	2021	Proposed an energy-efficient framework for real-time heart monitoring using a hierarchical edge-fog-cloud architecture within the IoMT	Simulated data is used to evaluate the framework's performance, demonstrating significant energy savings and improved scalability	The framework optimizes energy consumption through distributed processing and dynamic resource allocation, ensuring low-latency, real-time monitoring	Implementation complexity, dependence on network quality, security and privacy concerns, and the need for real-world validation are notable limitations
Mohana et al. (2022)	2022	Discussed the application of IoT in the healthcare field, leveraging CNN for processing health-related data. The focus is on using IoT devices to collect health data and CNNs to analyze this data	Included vital signs like heart rate, blood pressure, glucose levels, and other relevant health metrics	Real-time monitoring, Improved diagnostic accuracy, Scalability, and Versatility	Computational Requirements, Data Quality Dependency, Integration Complexity, Privacy and Security Concerns
Faruqui et al. (2021)	2021	Introduced a hybrid deep CNN model designed for diagnosing lung cancer. The model integrates computed tomography scan data and data from wearable sensor-based IoMT devices	Used publicly available CT scan datasets, such as the LUNA16 and LIDC-IDRI databases, which provide annotated images for lung cancer diagnosis	The approach offers significant advantages in terms of diagnostic accuracy and real-time monitoring	The approach faced limitations such as computational complexity, data integration challenges, dependence on data quality, and scalability issues
Zhang et al. (2021)	2021	Proposed a hybrid algorithm combining CNN and LSTM networks for detecting arrhythmia using RR interval signals in the context of a 5 G-enabled IoMT	MIT-BIH Arrhythmia datasets, which include labeled arrhythmia events used for training and testing the model	Improved detection accuracy, Real-time monitoring, Scalability, and Comprehensive analysis	Computational demand, Dependence on data quality, Implementation complexity, and Validation on diverse populations
Chen et al. (2020)	2020	Proposes an adaptive hybridized CNN model for predicting chronic kidney disease within an IoMT platform. The model uses IoMT data to predict chronic kidney disease and support timely and accurate medical interventions	utilizes datasets containing medical records and health parameters relevant to chronic kidney disease	High prediction accuracy, Adaptive learning, Real-time monitoring, And Comprehensive analysis	Computational complexity, Dependence on data quality, Implementation challenges, and Validation across populations
Sampathkumar et al. (2022)	2022	Introduced a model that integrated IoMT with big data analytics, utilizing a Reflective Belief Design framework and a CNN-Metaheuristic Optimization Procedure. The model aimed to enhance data analysis and decision-making in medical applications	Used large-scale health data collected from IoMT devices, including various physiological signals, medical records, and other relevant data	Improved accuracy, Efficient big data handling, Enhanced interpretability, and Scalability	Computational complexity, Dependence on data quality, and Implementation Challenges

Table 5 (continued)

Refs.	Year	Methodology	Datasets Used	Advantages	Limitations
Ismail et al. (2020)	2020	Proposed a CNN-based model for analyzing regular health factors within the IoMT environment. The model is designed to process and analyze health data collected from IoMT devices to monitor patients' health status	Used datasets comprising regular health factors, including vital signs and physiological data, collected from IoMT devices	High accuracy, Real-time analysis, Scalability, and Improved feature extraction	Computational complexity, Dependence on data quality, Generalization to diverse populations, and Integration challenges
More et al. (2020)	2020	Proposed a CNN-based model specifically designed to reconstruct medical images within IoMT. The focus is on ensuring the security and integrity of the medical images during reconstruction	Included various types of medical images such as X-rays, MRIs, and CT scans	Enhanced Security, High reconstruction accuracy, Efficiency, and Versatility	Computational resources, Dependence on data quality, Implementation complexity, and Scalability concerns
Al-Sa'D et al. (2018)	2018	Developed and trained a DL model to compress vital signs data, such as heart rate, blood pressure, and body temperature. The model is designed to minimize the loss of information while achieving significant compression ratios	Used real-world vital signs data from publicly available datasets. This data includes various physiological signals collected from patients, which are used to train and test the DL model	High compression ratio, Energy efficiency, Improved performance, and Scalability	Model complexity, Dependence on data quality, and Implementation challenges
Raj et al. (2020)	2020	proposed a DL-based model for medical image classification within the IoMT framework. The model focused on optimizing feature selection to enhance classification accuracy	Included images of different medical conditions, such as X-rays, MRIs, and CT scans	High classification accuracy, Reduced computational complexity, Scalability, and Comprehensive evaluation	Dependence on quality of data, complexity of model training, real-time processing challenges, and generalization of diverse datasets
Cao et al. (2020)	2020	Proposed a hybrid DL model that combines data compression and classification to enhance data delivery efficiency in mobile health applications. The model is designed to compress health data to reduce transmission costs and classify the data for healthcare analysis	Used health-related datasets, which include physiological signals and medical records, to train and test the hybrid model	Enhanced efficiency, High classification accuracy, Energy savings, and Scalability	Model complexity, dependence on data quality, real-time processing challenges, and generalization of diverse data

sification. The AlexNet model, which was released in 2012, was the first CNN model to perform at the cutting edge on the ImageNet Large Scale Visual Recognition Competition (ILSVRC) dataset. The AlexNet model has five convolutional layers, three max-pooling layers, and three fully connected layers, with a total of 60 million parameters. To avoid overfitting, AlexNet employs dropout regularization and rectified linear units (ReLU) as activation functions (Raj et al. 2020). On the other side, the VGG-16 model, which was released in 2014, excelled at state-of-the-art performance on the ILSVRC dataset. The VGG-16 contains 138 million parameters and comprises 13 convolutional layers and 3 fully linked layers. The VGG-16 model adds max-pooling layers after every two convolutional layers and employs 3x3 convolutional filters throughout the network. Dropout regularization and ReLU activation functions are also used by VGG-16 (Raj et al. 2020). Due to the deep architecture of the VGG-16 model and the usage of convolutional filters smaller than the AlexNet model, the VGG-16 model can learn more complicated features and perform better on image recognition tests. Nevertheless, the VGG-16 requires more computer power and training time because of its enormous number of parameters (Raj et al. 2020).

The evolution of DL and the development of CNN models for computer vision tasks are both directly attributable to AlexNet and VGG-16. The following are some examples illustrating the usage of VGG-16 and AlexNet models in image classification functions, along with the outcomes:

- On a dataset of dog and cat photographs, the VGG-16 and AlexNet models were trained to classify the images as either a dog or a cat. The dataset had 25,000 photographs in total, with 12,500 images in each class. After training, the accuracy of the VGG-16 model was 94.5%, compared to 93.8% for the AlexNet model (Cao et al. 2020).
- To identify the species of the flowers, the VGG-16 and AlexNet models were trained on a collection of flower photos (Raj et al. 2020). There were a total of 80 photos in each of the 17 categories of flowers in the dataset. After training, the accuracy of the VGG-16 model was 85.8%, compared to 81.1% accuracy for the AlexNet model (Cao et al. 2020).
- Chest X-rays can be used to determine whether or not a patient has pneumonia. Therefore, a dataset of chest X-ray images was used to train the VGG-16 and AlexNet models (Al-Sa'D et al. 2018). 5856 chest X-ray images were included in the dataset, of which 2682 were determined as pneumonia and 3174 as normal. After training, the VGG-16 model's accuracy was 91.5%, compared to 88.7% accuracy for the AlexNet model (Raj et al. 2020).

Table 5 provids a comprehensive overview of various research studies integrating IoMT with advanced ML and DL techniques to enhance healthcare applications. Each entry in the table summarizes key aspects of these studies, including their proposed models, methodologies, datasets utilized, advantages, and limitations. The studies span a range of applications, from real-time heart monitoring and chronic kidney disease prediction to lung cancer diagnosis and arrhythmia detection. The methodologies often leverage hierarchical architectures combining edge, fog, and cloud computing and employ advanced algorithms such as CNN, LSTM networks, and hybrid models. Datasets used in these studies include both simulated and real-world data, covering vital signs, physiological signals, and medical images. The advantages highlighted in these studies include improved accuracy, real-time monitoring, energy efficiency, and scalability. However, common limitations involve computational complexity, dependence on data quality, implementation challenges, and the need for real-world validation. This table underscores the potential and challenges of integrating IoMT with advanced analytics to revolutionize healthcare delivery and outcomes.

3 Blockchain with IoMT

A blockchain is a decentralized digital ledger technology that makes transactions safe and open. It is a database or ledger that is kept on a network of computers, with a copy of the ledger on each computer in the network (Garg et al. 2020; Rahmani et al. 2022). A blockchain's data is organized in blocks linked to one another in a sequential chain. Each block includes a collection of transactions and a distinct digital signature known as a "hash" that uniquely identifies the block (Jolfaei et al. 2021). The blockchain is an immutable and unchangeable record of all transactions that have ever taken place on the network. Once a block is added to the blockchain, it cannot be changed or removed. This makes blockchain technology perfect for use cases where security and transparency are crucial, such as cryptocurrency transactions, supply chain management, and voting systems (Jolfaei et al. 2021). The interconnected network of sensors, wearables, and medical equipment known as the IoMT is used to collect and send patient health data (Jolfaei et al. 2021). There are many ways in which blockchain technology can be applied to IoMT, as follows:

- **Safe data sharing**: Blockchain technology can allow patients, researchers, and various healthcare providers to share data in a decentralized and secure manner (Garg et al. 2020). It is feasible to guarantee that data is not changed or tampered with in any manner and that all parties have access to the same accurate information by using a distributed ledger (Jolfaei et al. 2021; Salunkhe 2023).
- **Authentication of medical equipment**: Blockchain technology can be used to confirm the reliability and integrity of medical devices. This is significant for implantable medical devices since security and dependability are essential (Sun et al. 2019; Singh and Das 2023).
- **Management of patient consent**: With the help of blockchain, people may remain in charge of their medical information and decide who has access to it. Also, the technology can ensure patients' permission is secured before sharing their data with any third parties (Rajput et al. 2019).
- **Supervision of drug supply chain**: From the manufacturer to the consumer, the complete pharmaceutical supply chain may be tracked and verified using blockchain technology (Garg et al. 2020). This can help ensure that patients obtain safe and effective treatments while preventing counterfeit products from entering the market (Singh and Das 2023).

Blockchain technology offers significant advantages for IoMT applications, particularly in enhancing the IoMT ecosystem's security, privacy, and interoperability, making it simpler for healthcare providers to share data, provide individualized care, and improve patient outcomes (Myrzashova et al. 2023; Dhasarathan et al. 2023). Its decentralized nature and robust security features make it an ideal choice for managing sensitive medical data and ensuring reliable data exchanges among healthcare providers. The main significant justifications for recommending blockchain technology for IoMT are as follows:

- **Improving security and privacy** (Jolfaei et al. 2021; Hegde and Maddikunta 2023):
  - **Immutable records**: Blockchain technology provides an immutable ledger for recording transactions. In the context of IoMT, this ensures that medical data remains tamper-proof and secure from unauthorized modifications, enhancing data integrity.
  - **Decentralization**: Unlike centralized systems, blockchain operates on a decentralized network of nodes, reducing the risk of single points of failure and making it more resilient to hack or manipulate.
  - **Privacy protection**: Blockchain can facilitate secure data sharing among healthcare providers while preserving patient privacy through techniques like zero-knowledge proofs and encrypted transactions.
- **Enhancing data interoperability** (Khan et al. 2022):
  - **Standardized protocols**: The IoMT ecosystem is highly fragmented, with various medical systems and devices that might not be compatible. A single data format and standard for data sharing can be established with blockchain, simplifying interoperability across multiple devices and systems.
  - **Smart contracts**: These are self-executing contracts with the terms of the agreement directly written into code. They can automate data exchange processes, enforce compliance with predefined rules, and streamline workflows across different healthcare entities.
- **Transparent and auditable transactions** (Musamih et al. 2022):
  - **Traceability**: Every transaction recorded on a blockchain is transparent and can be traced back to its origin. This feature is particularly useful in medical supply chains, clinical trials, and patient consent management.
  - **Auditability**: Blockchain provides a clear audit trail of all transactions, making it easier for healthcare providers and regulators to verify the authenticity and accuracy of medical records and data exchanges.
- **Cost reduction and efficiency** (Rahmani et al. 2022):
  - **Reduced administrative overhead**: By automating processes and eliminating intermediaries through smart contracts, blockchain can significantly reduce administrative costs and improve the efficiency of healthcare operations.
  - **Faster transactions**: Blockchain can facilitate faster and more secure transactions, such as insurance claims processing, reducing delays and improving the overall patient experience.
- **Improving data sharing**: Researchers, healthcare professionals, and patients may share

data transparently and safely thanks to blockchain technology(Hegde and Maddikunta 2023). It is feasible to guarantee that data is not changed or tampered with in any manner and that all parties have access to the same accurate information by using a distributed ledger (Garg et al. 2020).

- **Enhancing patient outcomes**: Blockchain gives people control over their medical data and access permissions. Better patient outcomes may result from increased patient engagement and more individualized care made possible by this (Khan et al. 2022).
- **Managing drug supply chain**: From the manufacturer to the consumer, the complete pharmaceutical supply chain may be tracked and verified using blockchain technology (Garg et al. 2020). This can help ensure that patients obtain safe and effective treatments while preventing counterfeit products from entering the market.
- **Enhancing regulatory compliance**: The secure and open transmission of sensitive patient data is required by laws like the U.S. Health Insurance Portability and Accountability Act (HIPAA), General Data Protection Regulation (GDPR), and the Drug Supply Chain Security Act, which blockchain can assist healthcare providers and pharmaceutical corporations in complying with (Jolfaei et al. 2021).

Its decentralized nature and robust security features make it an ideal choice for managing sensitive medical data and ensuring reliable data exchanges among healthcare providers. However, in scenarios where data security is less critical, or the scale of deployment is small, alternative techniques like traditional databases and cloud-based solutions may be more suitable. By carefully considering the specific requirements and constraints of each IoMT application, the most appropriate and effective technology can be selected to optimize performance, security, and cost-efficiency.

3.1 Network model

Patients, healthcare providers, and researchers can all share confidential data with other stakeholders in the healthcare ecosystem using a blockchain-based architecture for the IoMT (Khan et al. 2022). A high-level breakdown of the essential elements of an IoMT architecture based on blockchain is provided below:

- **Medical gadgets and sensors**: These are the ones that gather and communicate information about a patient's health. These could include medical sensors, implanted tech, and wearable tech (Khan et al. 2022; Rafique et al. 2023).
- **Gateways**: Devices are responsible for gathering data from sensors and medical devices and transmitting it to the blockchain network (Rafique et al. 2023). Tablets, cellphones, and other portable devices can all be considered gateway devices (Khan et al. 2022).
- **Blockchain network**: A distributed ledger that holds patient health data and facilitates safe and open data sharing amongst many stakeholders (Raj and Prakash 2023). Depending on the use case, the blockchain network may be public or private (Khan et al. 2022).
- **Smart contracts**: Self-executing contracts are kept on the blockchain network. They can automate the performance of specific tasks, like confirming patient permission and processing payments (Khan et al. 2022; Raj and Prakash 2023).
- **Decentralized applications**: Applications that run on top of the blockchain network.

They can be used to execute smart contracts, access patient health data, and carry out other operations (Khan et al. 2022).
- **Data analytics and AI tools**: They can be used to evaluate patient health data and produce insights that can be utilized to guide clinical decisions (Ellouze et al. 2020; Qu et al. 2022).
- **User interfaces**: They give various stakeholders an easy method to communicate with the BIoMT system. Web applications, mobile apps, and other user-facing tools may be among them (Egala et al. 2021).

By establishing a safe, decentralized ecosystem, the BIoMT communication environment network model revolutionizes healthcare (Raj and Prakash 2023). It makes data transmission and communication between medical devices, healthcare professionals, and patients effortless (Taherdoost 2023). Blockchain protects against manipulation while ensuring data integrity, privacy, and transparency, enabling real-time health data transmission. This revolutionary healthcare system revolutionizes, empowers individuals, and has great potential for the future, as shown in Fig. 5.

Ultimately, a BIoMT architecture could make it possible for many players in the healthcare ecosystem to share data in a secure, open, and decentralized manner (Khan et al. 2022). For healthcare businesses looking to use emerging technologies to improve their operations, it can help to improve patient outcomes, cut costs, and boost regulatory compliance (Khan et al. 2022).

3.2 The proposed BIoMT

This system is a blockchain-based architecture To improve the security, privacy, and interoperability of medical IoT devices (Jin et al. 2021). The system addresses major issues

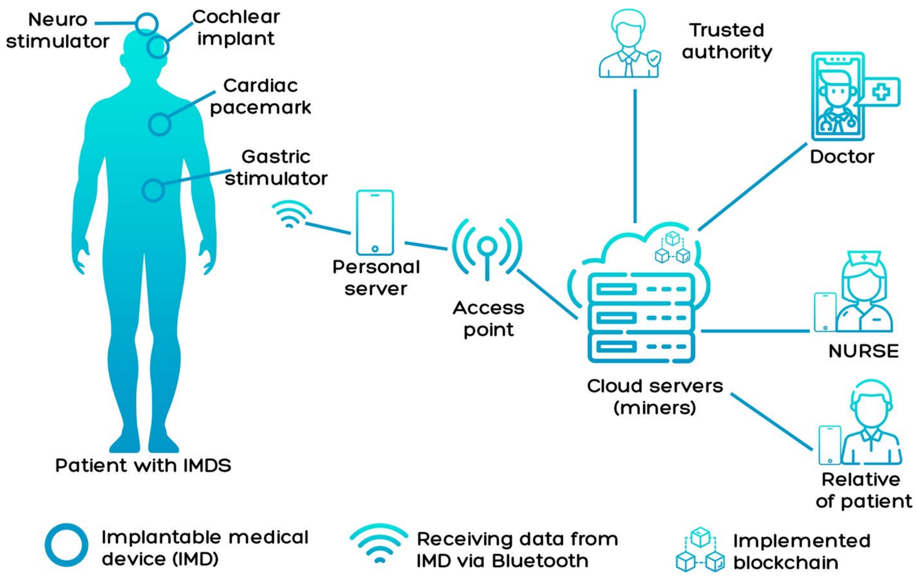

Fig. 5 BIoMT communication environment network model

the healthcare sector is now dealing with, including data fragmentation, interoperability, and security (Indumathi et al. 2020; Ellouze et al. 2020). The BIoMT architecture is built on a blockchain network that enables secure, open sharing of medical IoT data. The data, blockchain, and application layers comprise the system's three levels that make up the system (Kumar et al. 2023; Egala et al. 2021), as follows:

- **Data layer**: The sensors, IoT medical devices, and other data sources that produce patient health data comprise the data layer. Medical data from various sources must be gathered and processed by the data layer (Jin et al. 2021).
- **Blockchain layer**: This layer provides the BIoMT system's foundation. Using distributed ledger technology, the blockchain network safely and openly stores and distributes medical data (Jin et al. 2021). The blockchain layer is in charge of developing cryptography and consensus processes to guarantee the accuracy and security of medical data (Indumathi et al. 2020).
- **Application layer**: It is responsible for the BIoMT system's user interface. This layer enables access to and analysis of medical data stored on the blockchain network by healthcare providers, researchers, and other stakeholders (Kumar et al. 2023). Smart contracts, data analysis tools, and other applications that can use the medical data stored on the blockchain network are all implemented at the application layer (Indumathi et al. 2020).

The integration of the IoMT and blockchain technology has been increasingly explored to enhance security, data integrity, and efficiency in healthcare systems. IoMT refers to the network of interconnected medical devices that collect and transmit health data, while blockchain offers a decentralized and immutable ledger for secure data management. Recent studies have proposed various frameworks and models combining these technologies to address challenges such as data security, privacy, scalability, and interoperability. Table 6 provided an overview of these studies, highlighting their methodologies, datasets used, advantages, and limitations.

Compared to conventional medical IoT systems, the BIoMT system has various advantages. The system's use of blockchain technology enables secure, open, and interoperable medical data sharing, boosting healthcare delivery's effectiveness and efficiency (Indumathi et al. 2020; Al-Otaibi 2022). Because medical information is kept on a decentralized, impervious ledger that can only be viewed by authorized users, the technology also provides improved security and privacy (Kumar et al. 2023). The BIoMT system is additionally intended to be adaptable and scalable, enabling healthcare companies to incorporate new data sources and applications as required (Egala et al. 2021).

4 Cloud computing with IoMT

A pay-as-you-go method of accessing computing resources, including servers, storage, and applications, over the internet is known as cloud computing (Tahir et al. 2019). In the case of cloud computing, users can rent these resources from cloud service providers who look after and administer the infrastructure rather than hosting and managing their physical infrastructure (Khan and Algarni 2020; Awad et al. 2022). Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS) are just a few examples of the several kinds of cloud computing services available (Rahman and Hossain 2021; Bharati et al. 2021). IaaS gives customers access to virtualized computing resources like storage and virtual machines, whereas PaaS gives them a platform to create, deploy, and manage their applications. On the other hand, SaaS offers users online access to software programs (Rahman and Hossain 2021).

Scalability, flexibility, and cost-effectiveness are just a few advantages of cloud computing. Without purchasing and maintaining their physical infrastructure, customers may scale up or down their computer capabilities with cloud computing to meet their changing needs (Tahir et al. 2019; Khan and Algarni 2020). Moreover, cloud computing improves mobility and collaboration by enabling users to access their data and applications from any location with an internet connection (Rahman and Hossain 2021). Cloud computing has specific difficulties and worries, including data security and privacy issues, vendor lock-in, and potential service interruptions or outages (Pustokhina et al. 2020). Before making a choice, customers should, like with any technology, carefully consider their needs and the advantages and disadvantages of cloud computing (Darwish et al. 2019).

The IoMT, a rapidly growing industry, uses linked medical equipment and sensors to collect and deliver data for healthcare purposes (Bharati et al. 2021). Cloud computing is crucial to the IoMT ecosystem because it provides a scalable and adaptable infrastructure for storing, processing, and analyzing the vast volumes of data produced by these devices (Li et al. 2021; Rahman and Hossain 2021). One of the key benefits of cloud computing in the IoMT is that it makes it possible to monitor and analyze patient data on time, which can assist healthcare providers in spotting patterns, spotting anomalies, and making prompt and educated decisions (Tahir et al. 2019). IoMT systems can offer individualized suggestions and predictive insights by utilizing cloud-based analytics tools and ML algorithms, which can enhance patient outcomes (Khan and Algarni 2020; Bharati et al. 2021). The possibility for improved collaboration and information sharing across healthcare stakeholders, including patients, providers, and researchers, is another significant advantage of cloud computing in the IoMT (Awad et al. 2022). Healthcare firms may optimize workflows, boost communication, and speed up research and development using cloud-based systems to enable secure and effective data interchange and collaboration (Pustokhina et al. 2020; Moujahid et al. 2023). For IoMT applications, cloud computing is strongly recommended for the following reasons:

- **Scalability**: Cloud computing offers the ability to scale resources up or down based on demand. This is crucial for IoMT, which can generate large amounts of data that require storage and processing capabilities that can expand as needed.
- **Data storage and management**: IoMT devices generate massive amounts of data. Cloud platforms provide the infrastructure to store and manage this data efficiently, ensuring that it is accessible when needed.
- **Real-time data processing and analytics**: Cloud computing enables real-time processing of data collected from IoMT devices. This facilitates immediate analysis and decision-making, which is critical for timely medical interventions.
- **Cost efficiency**: By leveraging cloud services, healthcare organizations can reduce the costs associated with maintaining on-premises hardware and infrastructure. They pay only for the resources they use, which can be more cost-effective.

Table 6 Recent studies on blockchain with IoMT

Refs.	Year	Methodology	Datasets Used	Advantages	Limitations
Alkatheiri and Alghamdi (2023)	2023	Developed and proposed a blockchain-based security framework for IoMT devices	Utilized simulated data to test and validate the proposed blockchain-based security framework	Improved security, Data integrity, Efficient access control, and Attack resilience	Scalability concerns, Implementation complexity, Costly implementation, and Regulatory challenges
Qi et al. (2023)	2023	Propose and design a secure framework integrating blockchain technology with IoMT devices used for stress detection	Utilizes simulated data to test and validate the security and functionality of the proposed blockchain-based framework for stress detection	Improved accuracy in stress recognition	Scalability, Complex implementation, Regulatory compliance, and Resource intensive
Belhadi et al. (2023)	2023	Designed and proposed BIoMT-ISeg, a framework integrating blockchain technology to enhance the security and efficiency of medical image segmentation	Utilized simulated data and potentially medical imaging datasets to test and validate the proposed blockchain-based segmentation framework	High accuracy in segmentation, Improved security, Data integrity, and Efficient processing	Requires high computational resources
Albakri and Alqahtani (2023)	2023	Propose a comprehensive framework integrating blockchain for security, metaheuristics for optimization, and DL for intelligent data analysis in IoMT	Utilizes simulated data and potentially real-world healthcare datasets to validate the proposed framework's effectiveness and performance	Enhanced security, Optimized performance, Accurate analysis, and Improved healthcare services	Scalability concerns, Complex implementation, Resource intensive, and Regulatory compliance
Gupta et al. (2023)	2023	Comprehensive examination of blockchain integration in IoMT for healthcare, including theoretical frameworks, case studies, and practical implementations	The book likely discusses various datasets used in case studies and practical examples but does not focus on a specific dataset for the entire content	Security enhancement, Data integrity and authenticity, Efficient data management, and Enhanced patient care	Scalability issues, Complex implementation, Resource requirements, and Regulatory compliance
Kumar et al. (2022)	2022	Designed and proposed a combined blockchain and DL framework to enhance security and communication in industrial IoT networks	Utilized simulated data and potentially real-world industrial IoT network datasets to validate the proposed framework's effectiveness	Enhanced security, Efficient communication, Data privacy, and Real-Time monitoring	Scalability issues, Complex implementation, Resource intensive, and Regulatory compliance
Tomar and Tripathi (2023)	2023	Designed and proposed a framework combining blockchain technology with certificate-less aggregate signcryption to secure IoMT communications	Utilizes simulated data to test and validate the effectiveness and performance of the proposed scheme	Enhanced security, Efficient signcryption, Data integrity and privacy, and Reduced key management complexity	Potential issues with large-scale implementation, Resource intensive, and Regulatory compliance

Table 6 (continued)

Refs.	Year	Methodology	Datasets Used	Advantages	Limitations
Dai et al. (2021)	2021	Proposed a framework that combines blockchain with edge intelligence to improve security, data management, and decision-making in IoMT during the COVID-19 pandemic	Utilizes simulated data and potentially real-world COVID-19-related healthcare datasets to validate the proposed framework's effectiveness	Enhanced security, Efficient data management, Improved patient monitoring, and real-time decision-making	Edge devices' computational limitations, Scaling issues, Resource-intensiveness, and regulatory compliance
Chaudhury and Sau (2023)	2023	Designed and proposed a framework integrating blockchain technology with IoMT to improve the security and accuracy of breast cancer detection	Breast cancer imaging data, including mammograms	Accurate detection, Data integrity and authenticity, and Efficient data sharing	Scalability concerns, Complex implementation, Resource intensive, and Regulatory compliance
Rafique et al. (2023)	2023	Designed and proposed the Securemed framework, integrating blockchain with IoMT to secure data and ensure privacy	Utilizes simulated data to test and validate the Securemed framework's effectiveness and performance	Privacy preservation, Enhanced Security, Data Integrity, and Efficient Access Control	Scalability concerns, Complex implementation, Resource intensive, and Regulatory compliance
Zaabar et al. (2021)	2023	Designed and proposed the HealthBlock framework integrating blockchain to enhance data security, privacy, and management in healthcare	Utilizes simulated data and potentially real-world healthcare data to test and validate the HealthBlock framework's effectiveness	Avoid recorded security limitations in commonly used systems for smart healthcare	Implementation complexity in real-world scenarios
Dilawar et al. (2019)	2019	Explore blockchain integration to enhance IoMT security	None	Decentralization, Immutability, Automated processes, and Enhanced security	Scalability issues, Latency, and Complexity and cost
Dai et al. (2020)	2020	Theoretical examination of blockchain's role in improving data security and management during COVID-19	None	Enhanced security, Efficient data management, Real-time capabilities, and Transparency and trust	Scalability, Implementation complexity, and Regulatory compliance
Ellouze et al. (2020)	2020	Performed a technical analysis of blockchain features such as decentralization, immutability, and transparency and discussed their applicability to IoMT	None	Enhanced security and privacy, Data integrity, Transparency and traceability, and Interoperability	Scalability issues, Implementation complexity, Performance Overhead, and Regulatory compliance
Seliem and Elgazzar (2019)	2019	Presented the architecture of BIoMT, detailing the integration of blockchain with medical devices, data storage, and data sharing mechanisms	The evaluation of the proposed BIoMT framework is conducted using simulated data	Enhanced security and privacy, Data integrity, Transparency and trust, and Interoperability	Scalability challenges, Implementation complexity, Performance Overhead, and Regulatory and compliance issues

- **Improved collaboration and data sharing**: Cloud platforms enable seamless data sharing among healthcare providers, researchers, and patients. This enhances collaboration, leading to better patient outcomes and more efficient healthcare delivery.
- **Enhanced security and compliance**: Cloud providers invest heavily in security measures and compliance certifications. They offer advanced security features such as encryption, identity management, and regular security updates, helping healthcare organizations protect sensitive medical data and comply with regulations like HIPAA.
- **Disaster recovery and business continuity**: Cloud computing provides robust disaster recovery solutions. In the event of data loss or system failure, healthcare providers can quickly restore services, ensuring minimal disruption to patient care.
- **Remote monitoring and telemedicine**: Cloud computing enables remote monitoring of patients through IoMT devices. This supports telemedicine initiatives, allowing healthcare providers to monitor patient health in real-time and provide remote consultations and care.
- **Innovation and development**: Cloud platforms provide tools and services for developing and deploying IoMT applications. This accelerates innovation, enabling faster development cycles and quicker time-to-market for new healthcare solutions.

However, using cloud computing in the IoMT also presents several difficulties and worries, particularly regarding data security, privacy, and regulatory compliance (Khan and Algarni 2020; Darwish et al. 2019). It is crucial to ensure that the proper security measures are in place to protect patient confidentiality and adhere to all applicable laws and standards as IoMT systems produce and send sensitive health data (Komandur et al. 2022). It is crucial that IoMT stakeholders thoroughly consider their cloud computing alternatives and implement best practices for data management and security (Pustokhina et al. 2020; Darwish et al. 2019).

4.1 Architecture layers

A distributed computing paradigm, called edge computing, enables data to be processed and analyzed close to the origin of the data, as opposed to centralized data centers or the cloud (Moujahid et al. 2023). The edge computing architecture comprises different layers, each of which is responsible for specific duties and tasks (Xu and Ren 2022).

- **Device layer**: It is the foundational layer of the edge computing architecture and comprises interconnected gadgets, including sensors, cameras, and other IoT gadgets that gather information about the outside world. Since their computing and storage capacities are often constrained, these devices rely on the higher layers for processing and analysis (Pustokhina et al. 2020; Varshini et al. 2023).
- **Gateway layer**: Before sending data to the upper levels for additional analysis, the gateway layer is in charge of gathering, filtering, and preparing data from the device layer (Komandur et al. 2022). The device layer and the remainder of the edge computing architecture are connected through gateways, offering local connection, security, and network management functions (Xu and Ren 2022).
- **Edge computing layer**: The middle layer of the architecture, which is in charge of processing and interpreting data in real time and close to the data source. Small servers

or edge devices with more processing and storage capacity than the device layer often make up this layer (Pustokhina et al. 2020).

- **Cloud layer**: The uppermost tier of the edge computing architecture, which offers extra storage, processing, and analytic capabilities over and beyond those offered by the edge computing layer (Xu and Ren 2022). The cloud layer enables data sharing and collaboration across various edge nodes, as well as storing historical data and more complex analytics (Anusuya et al. 2023).

A strong and effective framework created to handle the problems brought on by the constantly growing network of connected devices is revealed by the layered architecture of edge computing (Varshini et al. 2023). In order to optimize data processing, increase efficiency, and enable new applications and services across a variety of domains, including smart cities, industrial automation, and healthcare, edge computing offers this layered architecture that delivers a scalable and distributed infrastructure, as shown in Fig. 6. By moving data processing and analysis closer to the data source, edge computing can lower latency, bandwidth consumption, and reliance on centralized data centers or the cloud (Gyamfi et al. 2023). Edge computing is a desirable alternative for many IoT and industrial applications because it can increase reliability, security, and scalability by dispersing computing resources over multiple layers (Pustokhina et al. 2020).

4.2 Technologies of cloud-enabled IoMT

The cloud-enabled IoMT uses several technologies to make it possible to gather, process, and analyze healthcare data (Pustokhina et al. 2020). Critical technologies in cloud-enabled IoMT include the following:

- A crucial component of cloud-enabled IoMT is cloud computing, which offers the services and infrastructure required to store and handle vast amounts of healthcare data. Scalability, flexibility, and cost-effectiveness provided by cloud computing make it a desirable option for many healthcare applications (Xu and Ren 2022).
- IoT devices, such as wearables, implantable medical equipment, and medical imaging equipment, are essential parts of a cloud-enabled IoT system. These gadgets gather and send patient data for processing and analysis on the cloud (Pustokhina et al. 2020).
- Another key technology in the cloud-enabled IoMT is edge computing, which enables real-time processing and analysis of healthcare data near the data source(Nair et al. 2023). Edge computing is beneficial for applications that need real-time analytics and decision-making since it can help to minimize latency and enhance overall system performance (Khan and Algarni 2020).
- In cloud-enabled IoMT, AI and ML algorithms are increasingly utilized to analyze healthcare data and produce insights that help guide patient care, diagnosis, and treatment. Healthcare data patterns and anomalies can be found using AI and ML, allowing for more precise and individualized treatment recommendations (Rahman and Hossain 2021).
- Blockchain technology can be deployed to improve data security, privacy, and integrity in cloud-enabled IoMT (Nair et al. 2023). Blockchain is a compelling alternative for applications that demand high levels of data security since it can help ensure that health-

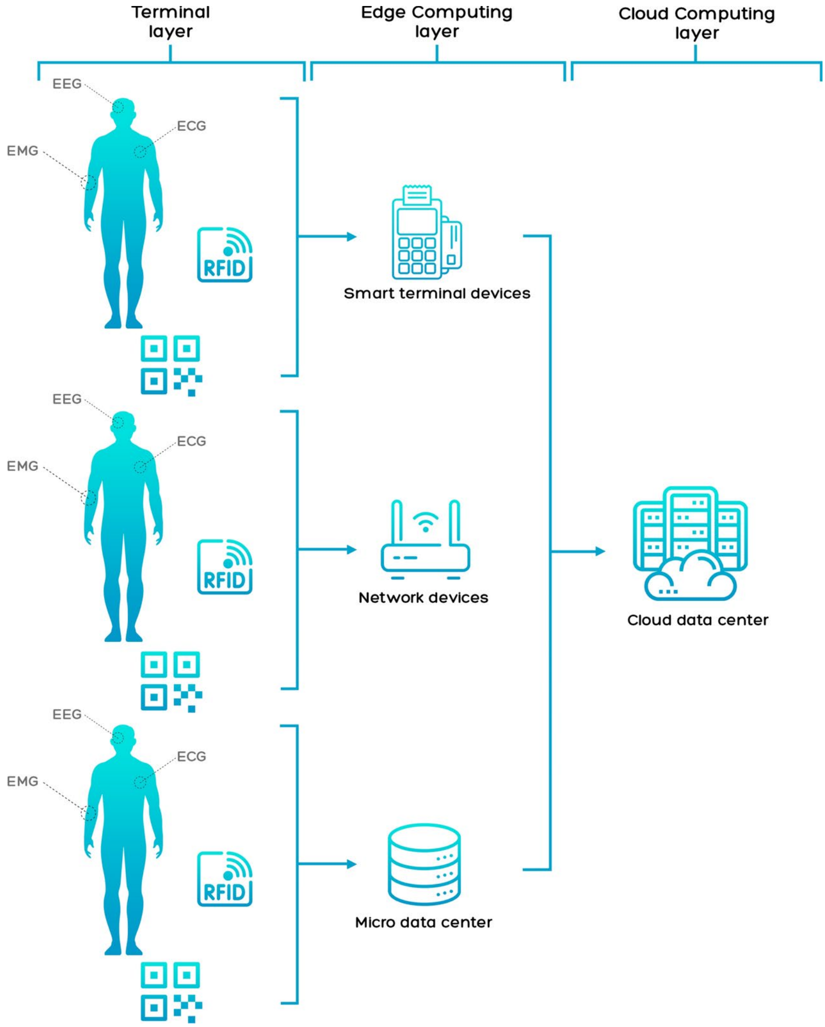

Fig. 6 The layered architecture of edge computing

care data is only accessible by authorized individuals and cannot be tampered with or altered (Alshehri and Muhammad 2020).

- IoMT powered by the cloud creates enormous volumes of healthcare data that must be handled and evaluated instantly (Nair et al. 2023; Gupta et al. 2022). This data can be analyzed using big data analytics techniques to produce insights that help guide healthcare decisions, improving patient outcomes and lowering costs (Deebak and Al-Turjman 2020).

Table 7 presented a comprehensive overview of recent studies focused on the integration of the IoMT and cloud computing. These studies explore various methodologies, ranging from literature surveys to proposing novel architectures and frameworks aimed at enhancing the

Table 7 Recent studies on cloud computing with IoMT

Refs.	Year	Methodology	Datasets Used	Advantages	Limitations
Sun et al. (2020)	2020	A comprehensive overview of integrating edge-cloud computing and AI within IoMT	Discusses data types involved in IoMT applications, such as patient records, real-time sensor data, and medical images	Enhanced data processing, scalability, and real-time analytics in medical applications	Implementation complexity, data security concerns, interoperability challenges, and cost considerations
Dang et al. (2019)	2019	A comprehensive survey of existing literature on IoT and cloud computing in healthcare	Compiles and analyzes data from various studies and applications	Broad overview of technologies, frameworks, and applications, highlighting challenges and opportunities	Lack of empirical data, broad scope, and rapidly evolving field
Tahir et al. (2019)	2019	Proposed a novel architecture for IoMT integrating fog and cloud computing for energy efficiency	Synthetic data generated through simulations	Energy efficiency, reduced latency, scalability, and dynamic resource management	Simulation-based evaluation, implementation complexity, and security/privacy concerns
Singh et al. (2021)	2021	Review of IoT and cloud computing integration in digital health	None	Insights into the benefits and challenges of IoT-cloud solutions in digital health	Lack of new empirical data, broad scope, and rapidly evolving field
Yıldırım et al. (2023)	2023	Proposed a framework integrating fog and cloud computing within IoMT for healthcare monitoring	Synthetic data generated through simulations	Reduced latency, energy efficiency, scalability, and improved data management	Simulation-based validation, implementation complexity, and Data privacy concerns
Alharbi et al. (2023)	2023	A four-tier architecture designed to optimize the placement and processing of IoMT applications across edge, fog, and cloud computing	Utilized simulation data to validate the proposed framework	Reduced latency by 73%, energy consumption up to 38%	Simulation-based validation, Resource constraints, and Scalability to diverse applications
Elhoseny et al. (2018)	2018	Involved integrating IoT devices for data collection and cloud infrastructure for data processing and storage	Various health-related data collected from IoT devices	Scalability, Improved healthcare services, and Real-Time processing	Data privacy and security, Dependence on network connectivity, and Implementation complexity
Deebak and Al-Turjman (2020)	2020	Introduces a smart mutual authentication protocol to enhance security in cloud-based medical healthcare systems	Employed simulated data to validate the proposed authentication protocol	Enhanced security, Mutual authentication, Efficient data protection, and Scalability	Complex implementation, Latency issues, Dependence on network stability, and Potential for security flaws
Darwish et al. (2019)	2019	Explored the integration of IoT and cloud computing to improve healthcare systems	The study is a review and does not use a specific dataset	Enhanced data management, Real-Time monitoring, Scalability, and Cost efficiency	Data security and privacy, Interoperability issues, Reliability and latency, and Complexity of implementation

Table 7 (continued)

Refs.	Year	Methodology	Datasets Used	Advantages	Limitations
Cui et al. (2021)	2021	Involved deploying IoT sensors and devices to collect real-time data on various health parameters such as cardiac speed and eye movement. This data is transmitted to cloud servers for processing and storage	Employed data collected from IoT sensors that monitor the aforementioned health parameters	Real-Time monitoring, Comprehensive data analysis, Scalability, and Improved health outcomes	Data security and privacy, Reliance on network connectivity, Complexity of implementation, and Latency issues
Ganesan et al. (2020)	2020	Involved using IoMT devices to collect medical data, which is then transmitted to cloud-based e-health services for analysis	Used MRI images of brain tumors to train and test the CNN model	Improved accuracy, scaleability, real-time monitoring, and cost-effectiveness	Data privacy and security, Dependence on network availability, Complexity in Integration, and High Computational Requirements
Cao et al. (2021)	2021	Focuses on developing a health monitoring system that integrates IoT devices with cloud computing to provide continuous health monitoring and analysis	Uses simulated health data collected from IoT sensors for system testing and validation	Real-Time monitoring, Scalability, Accessibility, and Cost-Effectiveness	Data Security and privacy, Dependence on network connectivity, Implementation complexity, and Latency issues

efficiency, scalability, and security of healthcare systems. The datasets utilized include both real-time data from IoT devices and synthetic data generated through simulations, highlighting the diversity in research approaches. Key advantages identified across these studies include improved data processing capabilities, reduced latency, energy efficiency, and enhanced scalability. However, common limitations such as implementation complexity, data security and privacy concerns, and the need for empirical validation are also noted. This table is a valuable resource for understanding the current landscape and identifying areas for further research in the domain of IoMT and cloud computing.

Overall, the numerous and quickly developing technologies used in cloud-enabled IoMT provide new and cutting-edge applications in healthcare (Gupta et al. 2022). To ensure the safe and efficient use of these technologies in healthcare, several issues, including data privacy, security, and regulatory compliance, must be adequately addressed due to the adoption of these technologies (Alshehri and Muhammad 2020).

5 IoMT security

As our reliance on technology increases, security in technology is a crucial issue that is getting increasingly significant (Papaioannou et al. 2022). The hazards of cyberattacks and data breaches have significantly increased with the spread of internet-connected devices, from smartphones to smart homes. People and organizations must, therefore, take action to secure their technology and safeguard their data (Radoglou-Grammatikis et al. 2021). Technology security is a broad topic with many facets. Protecting against unauthorized access is among the most crucial (Papaioannou et al. 2022). This problem may entail putting solid passwords, two-factor authentication, and other security procedures in place to guarantee that only authorized users can access sensitive data or systems. In addition, it involves protecting systems and devices from viruses, hacking attempts, and other dangers (Deebak et al. 2019; Yaacoub et al. 2020).

Data protection is a crucial component of technology security. Both critical corporate information and personal data are included in this (Ghubaish et al. 2020; Satamraju and Malarkodi 2021). Encrypting data in transit and at rest, frequently backing up data, and putting access controls to ensure that only authorized individuals may view or edit data are all examples of data security measures (Deebak et al. 2019; Paul et al. 2023). Technology security is crucial for maintaining the reliability of systems and data. This problem entails avoiding data manipulation, like putting audit trails into place, ensuring that software and hardware are current and secure, and keeping an eye on systems for any indications of unusual activity (Manimurugan et al. 2020). The network of networked wearables, sensors, and other healthcare technologies known as the IoMT is used to gather, transmit, and analyze patient data. These gadgets are revolutionizing healthcare, enabling more individualized and effective care (Radoglou-Grammatikis et al. 2021; Sharma et al. 2023). However, there are significant security issues that need to be addressed with the growth of the IoMT (Satamraju and Malarkodi 2021). Security in the IoMT is essential for the following reasons:

- **Safeguarding patient data**: It is one of the fundamental reasons why security is necessary for the IoMT. IoMT devices gather sensitive information, such as patient health data, medical histories, and treatment plans (Yaacoub et al. 2020; Sharma et al. 2023).

Identity theft, fraud, and other nefarious actions may result if this data gets into the wrong hands. It is critical to ensure the security of this data to preserve patients' privacy and maintain the reliability of the healthcare system (Manimurugan et al. 2020; Mohan et al. 2022).

- **Defending against cyberattacks**: The IoMT is susceptible to cyberattacks, just like any other networked system. Hackers may try to steal patient data from IoMT devices, obtain illegal access, or interfere with the devices' operation (Mohan et al. 2022). A successful cyberattack could have serious consequences, including compromised health care or patient injury. To stop these kinds of attacks, strong security measures like encryption and access controls are required (Deebak et al. 2019).
- **Preserving system integrity**: The integrity of the healthcare system is likewise put in danger by the IoMT. Inaccurate data transmission from a compromised IoMT device could result in incorrect diagnosis or therapy recommendations (Paul et al. 2023). Maintaining the integrity of the healthcare system and giving patients safe and effective care depends on IoMT devices being secure and operating properly (Manimurugan et al. 2020; Xu et al. 2023).
- **Regulatory compliance**: Healthcare providers and organizations must adhere to several rules and guidelines, including HIPAA and GDPR, to secure patient data and maintain the security of their systems (Ghubaish et al. 2020). To comply with these rules and requirements, the IoMT must have robust security measures in place (Radoglou-Grammatikis et al. 2021). Severe penalties and reputational harm may arise from failure to abide by these requirements.

5.1 Security properties

A system, program, or device's security properties are its traits or qualities that enhance overall security (Saheed and Arowolo 2021). While developing, putting into practice, or evaluating the security of a system, designers frequently take into account several crucial security properties (Bhushan et al. 2023), including:

- **Confidentiality**: The ability to keep data or information secret and shield it from unauthorized access. Measures like data masking, access limits, and encryption can help with this (Parah et al. 2020).
- **Integrity**: The guarantee that data or information has not been altered or changed in any unapproved manner. Digital signatures, data backups, and audit trails can help with this (Hasan et al. 2021).
- **Availability**: The promise that a system or program will be usable and available when required. Measures like redundancy, fault tolerance, and disaster recovery planning can help with this (Nguyen et al. 2021).
- **Authentication**: Verifying a user's or a device's identity when accessing a system or application. Passwords, fingerprints, and other security mechanisms like multi-factor authentication can be used to do this (Nguyen et al. 2021).
- **Authorization**: The process of allowing or denying access to a system or application depending on the identification and permissions of the person or device. Role-based access control, attribute-based access control, and required access control are a few strategies that can be used to accomplish this (Villegas-Ch et al. 2023; Saheed and Arowolo

2021).

- **Non-repudiation**: The assurance that a user or device can't dispute performing a certain activity, such as sending a message or carrying out a transaction. Measures like transaction logs and digital signatures can be used to accomplish this (Parah et al. 2020).
- **Auditability**: The capacity to track and monitor system behavior to recognize and look into security incidents. Log files, intrusion detection systems, and security information and event management technologies are some tools that can help with this (Parah et al. 2020).In general, security features are crucial for assuring authentication, authorization, non-repudiation, auditability, confidentiality, integrity, and availability of data and systems. Each system or application that demands security should consider these qualities and include them in its design (Saheed and Arowolo 2021).

5.2 Types of attacks

The IoMT, which offers remote monitoring, real-time patient data collecting, and improved patient care, has created new opportunities for the healthcare sector (Saheed and Arowolo 2021; Anusha et al. 2023). The increased connectivity and data sharing across medical systems and devices have created security risks for IoMT systems, nevertheless (Parah et al. 2020; Liaqat et al. 2020). The following are some of the most typical assault kinds in IoMT:

- **Distributed Denial-of-Service (DDoS) and Denial-of-Service (DoS)**: An attempt to turn off a system or network by flooding it with data or traffic (Wazid et al. 2019). In the case of IoMT, this would entail sending many requests or data packets to a medical system or device, making it crash or stop responding. While a DDoS attack is comparable, it is far more difficult to counter because it is launched from numerous sources (Saheed and Arowolo 2021).
- **Ransomware**: It encrypts a victim's data and demands money for the decryption key (Gupta et al. 2023). Ransomware is a sort of malware. In the case of IoMT, this can entail encrypting the patient data kept on a system or device for use by healthcare professionals, rendering it inaccessible unless they pay a ransom (Hasan et al. 2021; Liaqat et al. 2020).
- **Malware**: Any software intended to harm or exploit a system or network. Malware could be used in IoMT to steal patient data, interfere with medical equipment, or obtain illegal access to a network (Hasan et al. 2021).
- **Man in the Middle**: Attacks by a "Man in the Middle" include the attacker intercepting and changing the communications between two parties. Man in the middle could entail intercepting and changing patient data when sent between a medical device and a healthcare provider in the context of IoMT (Wazid et al. 2019).
- **SQL injection**: Using a weak application, malicious code is injected into a database in a SQL injection attack. In the context of IoMT, this can entail taking advantage of a weakness in the software of a medical device to access patient information without authorization or take over the device directly (Hasan et al. 2021).
- **Social engineering**: This term describes several attack methods intended to deceive or influence victims into disclosing private information or taking activities that jeopardize security. In the context of IoMT, this can entail pretending to be a medical professional or IT support staff to get access to a medical system or device (Parah et al. 2020).

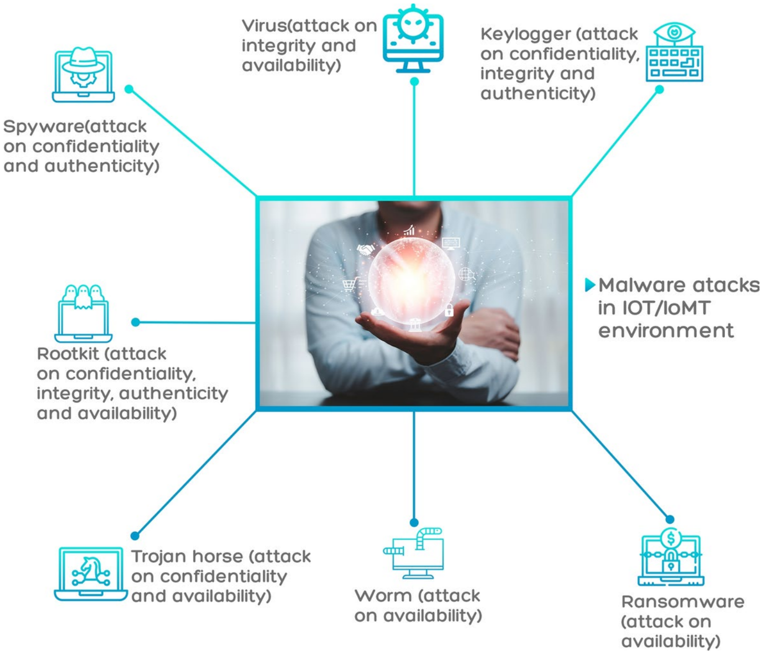

Fig. 7 Types of attacks occurred in IoMT systems

- **Insider threats**: Security dangers posed by staff members, subcontractors, or other dependable people with access to private information or computer systems (Wazid et al. 2019). In the context of IoMT, this might entail a worker purposefully stealing patient data or infecting a system with malware (Nguyen et al. 2021).

IoMT systems are subject to various assaults, including key-logger, spyware, rootkit, and malware attacks (Sun et al. 2019), as shown in Fig. 7. Malware undermines system security, spyware captures private data, rootkits allow unwanted access, and key-loggers track keystrokes. The integrity of the IoMT system and patient data are in danger due to these attacks (Nguyen et al. 2021; Wazid et al. 2019). Protection requires effective security measures, such as software upgrades and anti-malware programs (Hasan et al. 2021).

It's crucial to remember that these are just some of the most typical assault kinds in IoMT and that new attack techniques are continually being developed (Ghubaish et al. 2020). Healthcare providers and organizations must implement strong security measures, such as encryption, access controls, intrusion detection systems, and employee training and awareness programs, to reduce the risks of these attacks. Regular security audits and penetration testing can also aid in finding holes and addressing them before attackers take advantage of them (Nguyen et al. 2021).

6 IoMT application domains, challenges, limitations, and future development

The IoMT, a fast-developing field, is revolutionizing the delivery of healthcare. The IoMT is a medical software and hardware network allowing real-time data monitoring, analysis, and exchange. With the development of IoMT, medical professionals can remotely monitor patients, keep tabs on their health, and more accurately identify and treat ailments (Alshehri and Muhammad 2020; Al-Turjman et al. 2020). The future of IoMT is bright, and it has the potential to improve healthcare in many ways. Healthcare practitioners will be able to better care for patients, lower costs, and improve patient outcomes as IoMT devices become more widely used. The IoMT will be able to provide patients all over the world with truly tailored and efficient healthcare by utilizing cutting-edge technology like AI, ML, and blockchain (Din et al. 2019).

Healthcare is a fundamental aspect of life, yet modern healthcare systems are increasingly strained due to a steadily aging population and a corresponding rise in chronic illnesses. This escalating demand for resources, including hospital beds, doctors, and nurses, underscores the urgent need for a solution to alleviate pressure on these systems while maintaining high-quality care for at-risk patients. In recent years, the healthcare industry has recognized a growing interest in developing effective and practical medical care solutions. These innovations aim to assist hospitals and healthcare providers in streamlining diagnoses and enhancing patient care (Webber et al. 2022).

Human activity recognition (HAR) can be applied to determine worker presence, metatagging, and human counting, which is particularly useful for limiting the number of people in an area during pandemic conditions. By analyzing reflected and blocked wireless local area network (WLAN) signals, researchers achieved near-perfect accuracy using multi-class support vector machines (SVM) to differentiate user activities. On the other hand, employing wearable sensors enables HAR in both indoor and outdoor settings with fewer restrictions. Common sensors used for HAR include accelerometers, temperature sensors, radar, pressure sensors, stress sensors, and magnetic field sensors. A significant challenge in HAR is processing large volumes of sensing data with low latency, for which ML algorithms are particularly well-suited (Webber et al. 2022).

One of the core features of 5 G is network slicing, which enables the provision of a wide range of services, including those related to the Internet of Things (IoT). To support these services effectively, the 5 G network requires enhancements in network and communication capabilities, such as high throughput, dependability, and low latency. These improvements are essential for meeting the diverse and demanding requirements of IoT applications. This makes 5 G particularly well-suited to support HAR technologies, facilitating efficient processing and transmission of the large volumes of data generated by various sensors and WLAN signals (Dangi and Lalwani 2023).

Furthermore, the IoT has fueled investigations into transforming urban systems into "smart cities", enabled by technological advancements in computation, information and communication technology, IoT, robotics, and autonomous systems. The emergence of smart cities has led to the exploration of technologies that can be harnessed to more efficiently monitor and manage green stormwater infrastructure and nature-based solutions (NBS). These advancements are critical for helping cities adapt to climate change and deliver ecosystem services more effectively. Such interventions pave the way for sustainability, transforming cities into intricate social-ecological-technological systems where technology and environmental functionality dictate the necessary changes (Chen et al. 2023).

However, despite the advancements in healthcare technology, there remains a significant gap in comprehensive and systematic reviews that consolidate these developments, particularly in the context of the Internet of Things (IoT). The integration of IoT in healthcare has the potential to revolutionize patient monitoring, data collection, and overall care efficiency. Yet, there is a paucity of research that systematically examines the scope, implementation challenges, and outcomes of IoT applications in this field. This study aims to address this gap by providing a detailed systematic review of current IoT applications in healthcare, identifying the challenges and benefits, and offering insights into future directions. By focusing on this research gap, our study highlights the novelty and significance of systematically analyzing IoT's role in modern healthcare, thereby contributing valuable knowledge to both academic and practical domains (Osama et al. 2023).

6.1 Application domains

The IoMT spans various applications, transforming healthcare delivery and improving patient outcomes. The primary application domains of IoMT include:

- I-GCMS-IoMT based Medical Nursing System (Joyia et al. 2017).
- AR-IoMT based Smart Rehabilitation System (Dwivedi et al. 2022).
- IoMT based Kidney abnormality detection system using ultrasound imaging (Ma et al. 2020).
- IoMT Application for patient posture recognition using supervised learning (Manickam et al. 2022).
- Monitoring patient/ Remote patient monitoring (AlShorman et al. 2020).
- Decision making and home-based medical health monitoring system for neurologically disabled patients (Juneja et al. 2021).
- Autistic patient monitoring medical health care system using IoMT (Das et al. 2019).
- Smart medical nursing healthcare system for patients (Bendre and Khurge 2022).
- Remotely ECG monitoring system based on IoMT (Parvathy et al. 2021).
- BAKMP-IoMT: Secured and smart medical healthcare system (Dwivedi et al. 2022).
- EAI-IoMT based smart medical health band to monitor elderly people (Raza et al. 2022).
- I-GCMS based smart Hospital (Tang et al. 2019).
- SpO2: Monitoring of OSA (obstructive sleep apnea) diseased patient by heart rate variability (Haoyu et al. 2019).
- BACKM-EHA: Mobile electronic medical health care system based on IoMT (Wazid and Gope 2023).
- IoMT based Inexpensive cardiac arrhythmia management (ICarMa) system for cardiac patients (Khan et al. 2021).
- IoMT based MSSO-ANFIS medical healthcare monitoring system for heart disease prediction (Khan and Algarni 2020).
- IoMT based Medical Bot Michael for identification of carotid plaque risk factors (Thorsson et al. 2018).
- Secure-iGLU-IoMT based Glucose monitoring (Joshi et al. 2020).
- BioSenHealth 1.0-IoMT based Heart-rate monitoring system (Nayyar et al. 2019).

- Hand hygiene monitoring (Srigley et al. 2015).
- Depress-DCNF-IoMT used for Depression and mood monitoring (Kumar et al. 2022).
- Parkinson's disease monitoring (Zhang et al. 2022).

6.2 IoMT applicability and solutions for inventory management and supply chain optimization

The IoMT integrates medical devices and sensors with internet connectivity to enable real-time monitoring and data exchange. Beyond enhancing patient care, IoMT can significantly improve inventory management and supply chain optimization in healthcare. By addressing challenges such as stockouts, overstocking, and resource allocation, IoMT provides innovative solutions to streamline operations and improve efficiency.

6.2.1 IoMT for inventory management

- **Real-Time Tracking**: IoMT utilizes sensors and RFID tags attached to medical supplies and equipment to enable continuous monitoring and real-time data collection. This allows healthcare providers to maintain optimal stock levels, reducing the risk of both stockouts and overstocking. Automated systems can update inventory levels immediately, ensuring that supplies are always available when needed (Lee and Lee 2020).
- **Automated Replenishment**: IoMT-enabled cabinets and shelves equipped with weight sensors monitor stock levels and automatically trigger replenishment orders when inventory falls below a predetermined threshold. Predictive analytics analyze historical usage patterns to forecast future demand, ensuring timely restocking of critical supplies and minimizing shortages (Rani et al. 2023).
- **Temperature and Environmental Monitoring**: IoMT devices are essential for managing the cold chain by continuously monitoring temperature and humidity levels in storage areas. These devices ensure that temperature-sensitive items like vaccines and medications are stored optimally. Alerts are sent out if any deviations occur, allowing immediate corrective actions to maintain the integrity of stored items (Perera et al. 2015).

6.2.2 IoMT for supply chain optimization

- **Supply Chain Visibility**: IoMT provides end-to-end visibility of the supply chain through real-time tracking of goods from manufacturers to end-users. Integrating blockchain technology ensures secure and tamper-proof records of every transaction and movement within the supply chain, enhancing transparency and traceability (Tokkozhina et al. 2023).
- **Efficient Resource Allocation**: IoMT data enables dynamic routing and scheduling of deliveries based on real-time conditions, optimizing resource utilization and reducing costs. Real-time tracking of delivery vehicles and medical equipment ensures efficient use of assets, minimizing idle time and improving operational performance (Marinagi et al. 2023).

- **Demand Forecasting and Inventory Optimization**: Combining IoMT data with AI and ML algorithms allows healthcare providers to predict demand patterns more accurately, optimizing inventory levels and reducing wastage. Real-time data sharing between suppliers and healthcare providers enhances coordination, ensuring that inventory aligns with actual demand and reducing the risk of overstocking or stockouts (Tahir and Khan 2023).

6.3 Challenges

While the IoMT has many advantages for both patients and healthcare professionals, there are also considerable drawbacks to this technology (Amin et al. 2019; Ray et al. 2020; Quy et al. 2022). The following are a few of IoMT's main obstacles:

- **Data privacy and security**: are major concerns regarding IoMT devices, which are used to collect, transmit, and store sensitive patient data (Al-Turjman et al. 2020). These devices may produce data that contains personal and medical information that must be protected. Data breaches, illegal access, or data misuse can pose a significant risk and may lead to identity theft, medical fraud, or patient harm (Navaz et al. 2021). Inherent security and privacy risks are the biggest obstacles to applying IoMT in healthcare systems (Shakeel et al. 2023; Majeed et al. 2023). Surveys have repeatedly shown that healthcare institutions are concerned with potential data breaches and unauthorized access to private patient data (Sivarajah et al. 2023). Ensuring strong encryption, authentication, and data access restrictions for a wide range of connected devices is a challenging task that requires ongoing monitoring and updates to keep up with evolving cybersecurity threats (Majeed et al. 2023).
- **Interoperability**: It might not be easy to integrate and transfer data across many systems because different manufacturers frequently create IoMT devices and may have several communication protocols or data formats (Ray et al. 2020). The fragmentation of healthcare data from this interoperability problem could lead to incorrect diagnosis or inefficient therapies (Navaz et al. 2021; Manogaran et al. 2018).
- **Regulatory compliance**: Because the IoMT market is expanding quickly, many products are not overseen by the same regulations as conventional medical devices (Al-Turjman et al. 2020). Making IoMT devices abide by legal specifications for data privacy, security, and safety is crucial. Regulator compliance that is inaccurate or insufficient may have negative legal and financial repercussions (Chen et al. 2021). To guarantee patient safety and data integrity, it is crucial to comply with healthcare legislation and standards (Si-Ahmed et al. 2023). Recent surveys have uncovered issues with using IOMT systems while following regulatory regulations (Sivarajah et al. 2023). Healthcare providers and technology developers may feel uncertain due to the constantly shifting and occasionally hazy regulatory environment, which could slow down the implementation of IOMT in healthcare systems (Singh et al. 2023).
- **Reliability**: The measurements and readings of IoMT devices must be accurate and reliable. Even small mistakes or inaccuracies could significantly impact patient health outcomes. These devices' dependability also relies on their capacity to function properly under various environmental factors, including temperature and humidity (Chen et al. 2021).

- **Cost**: IoMT devices can be pricey, and healthcare providers may incur a high upfront cost when purchasing them. Also, some providers may find it difficult to afford the continuous expenditures of maintaining and improving the equipment (Navaz et al. 2021). IOMT has the potential to lead to better healthcare outcomes, but the upfront expenditures and annual maintenance fees can be high (Singh et al. 2023; Majeed et al. 2023). The cost expense of deploying and operating IOMT devices is a worry for healthcare institutions, especially smaller ones, according to surveys (Rani et al. 2023). The widespread use of IOMT in some healthcare settings may be hampered by these financial restrictions (Khan et al. 2023).
- **User acceptance**: IoMT faces a major barrier in gaining user acceptance. These gadgets could not be well-known to patients and healthcare professionals, and they might not trust them, which could result in poor adoption rates. Campaigns for education and awareness can help alleviate these worries and highlight the advantages of IoMT devices (Malamas et al. 2021). In order for medical staff and professionals to use IOMT efficiently, healthcare systems must provide them with the necessary training (Rani et al. 2023). Surveys have revealed that user adoption and training remain a major barrier since some healthcare professionals may be reluctant to adopt new technology, resulting in underuse or inefficient use of IOMT devices (Singh et al. 2023).
- **Data Overload and Data Management**: Healthcare practitioners must efficiently process and interpret the massive amounts of data that are generated by the implementation of IOMT devices (Khan et al. 2023; Si-Ahmed et al. 2023). Since polls show that healthcare organizations struggle to handle, analyze, and derive useful insights from the influx of real-time data, dealing with this data deluge creates serious difficulties (Shakeel et al. 2023; Alhaj et al. 2022). Medical professionals may become overloaded with information as a result, which could cause them to miss important patient data (Si-Ahmed et al. 2023).

The IoMT offers a lot of potential to enhance healthcare outcomes, but there are also a lot of issues that need to be resolved (Ray et al. 2020). The most important difficulties facing IoMT are data privacy and security, interoperability, regulatory compliance, reliability, cost, and user adoption (Manogaran et al. 2018). To overcome these obstacles and guarantee that IoMT devices are used safely and morally, healthcare practitioners, manufacturers, and policymakers must collaborate (Chen et al. 2021).

The integration of the IoMT in healthcare presents numerous ongoing challenges that the research community is actively addressing. One significant area of concern is ethical considerations. The use of IoMT raises substantial ethical issues related to data ownership, patient consent, and the responsible use of patient-generated data. These issues include the ethical implications of data collection, sharing, and usage in research, as well as the potential risks of introducing biases or discrimination through these processes. Addressing these challenges requires ensuring patient privacy, obtaining informed consent, and implementing robust data protection measures. Additionally, it is essential to establish transparent and equitable practices to prevent data misuse and to foster trust in IoMT technologies (Pournik et al. 2023).

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Legal challenges also pose significant obstacles to the widespread adoption of telemedicine and IoMT solutions. A study from the Department of Health Services Management at North Tehran Branch, Islamic Azad University, Tehran, Iran, highlights several legal challenges in telemedicine implementation. These include the incompatibility of existing laws with the electronic world, the ambiguity of professional and legal responsibilities, the lack of a legal mechanism for authenticating service providers, and the absence of a legal mechanism for authenticating service recipients. The study emphasizes that the failure to update laws in line with electronic advancements hampers telemedicine execution. Implementing electronic prescriptions, orders, and referrals requires a national infrastructure that supports electronic signatures and their legal recognition. Much of the existing healthcare legislation, such as those governing doctors' documents and patient information, remains rooted in the paper world, creating obstacles for telemedicine. The study also notes that using two-factor passwords alone cannot provide adequate security for service deliverers due to the high risk of hacking (Hosseini et al. 2024).

Another ongoing challenge is power consumption. High power consumption and the reliance on battery-powered devices necessitate frequent replacements, which can be impractical and costly. Ensuring that IoMT devices are energy-efficient while maintaining their functionality is crucial for their long-term viability. Researchers are working on developing low-power consumption technologies and improving battery life to address this issue (Niu et al. 2024).

The integration of the IoMT into healthcare systems introduces significant challenges, particularly in the realm of analytics on streaming datasets. The diverse nature of IoMT systems results in highly dimensional data collection, increasing the complexity of processing and analysis. Additionally, the presence of unstructured data types such as text, images, and symbols from various components poses challenges in analysis and modeling (Alturki et al. 2024).

Transmitting this data over wireless networks exposes it to noise, loss, and security threats, potentially affecting data quality and reliability. Therefore, it is crucial to address these issues when dealing with IoMT data. The heterogeneity of IoMT extends to protocol diversity, including ZigBee, RFID, LoRa, and TCP/IP, which further complicates data acquisition, manipulation, analysis, and modeling (Alturki et al. 2024).

One of the critical challenges in IoMT analytics is handling the volume and velocity of streaming data. Real-time analytics on such data requires robust decision-making, security, and prediction capabilities to ensure timely and accurate insights. The communication of information among heterogeneous protocols is essential for making informed decisions effectively. However, interoperability challenges may arise due to standard differences, making cooperation between protocols difficult. For instance, an IP-based hub may face difficulties in exchanging data directly with a Bluetooth-based IoT-Connected Inhaler, potentially necessitating the use of a converter (Alturki et al. 2024).

By highlighting these challenges, this study underscores the complexity of integrating IoT in healthcare and the importance of addressing ethical, legal, and technical issues to realize the potential of IoMT technologies fully. The ongoing efforts to overcome these challenges are vital for the successful implementation and adoption of IoT in healthcare, ultimately leading to improved patient outcomes and more efficient healthcare systems.

6.4 Limitations

Although there are some limits to this technology, the IoMT can potentially transform healthcare delivery and enhance patient outcomes (Quy et al. 2022). The following are some of the main IoMT limitations:

- **Restricted accessibility**: Even though IoMT devices are increasing, not all patients can use the technology. The capacity of some patients to connect to I0MT devices may be hampered by a lack of financial resources or the fact that they live in remote places with poor internet access. This restriction can widen the digital divide, with patients with less access to IoMT devices potentially receiving worse care than those who do (Vishnu et al. 2020; Hettikankanamage and Halgamuge 2021).
- **Technical complexity**: For patients and healthcare professionals to use IoMT devices appropriately, they may need to undergo specialized training. Its intricacy raises the possibility of data transmission or data gathering errors, which could result in misdiagnoses or unsuccessful therapies (Pal et al. 2018).
- **Minimal clinical evidence**: IoMT devices can potentially improve patient outcomes, but there is little clinical proof. Some research has examined the long-term advantages of IoMT devices; most have concentrated on short-term results. Their acceptance by healthcare providers, payers, and politicians may be constrained by this absence of evidence (Khan and Algarni 2020).
- **Ethical issues**: IoMT devices raise ethical issues, notably data security and privacy. People could be reluctant to divulge their private information to healthcare practitioners and independent businesses that design and manage these devices. The usage of IoMT devices and their potential advantages may be constrained by this reluctance (Khan and Algarni 2020).
- **Regulatory challenges**: In nations with various regulatory frameworks, regulatory control for IoMT devices can be complicated and fragmented. The lack of a uniform regulatory framework can hinder the development and use of IoMT devices by confusing manufacturers and healthcare practitioners (Pal et al. 2018).
- **Integration with current systems**: Because existing healthcare systems are so disjointed, integrating IoMT devices with them might be difficult (Hettikankanamage and Halgamuge 2021). The usefulness of IoMT devices may be constrained by the incompatibility of those devices with electronic medical records or other methods for managing healthcare data (Khan and Algarni 2020).
- **Issues with Interoperability**: There are many different types of IOMT devices, each with their own set of protocols, communication standards, and data formats (Alhaj et al. 2022). A significant obstacle to seamless data transmission and interoperability across various devices and healthcare systems is this lack of standardization (Khan et al. 2023; Sivarajah et al. 2023). According to surveys, integrating different IOMT devices into current healthcare infrastructures is frequently difficult, creating data silos and impeding the thorough study of patient data (Shakeel et al. 2023).
- **Analytics on Streaming Datasets**: Analytics on streaming datasets in healthcare face several challenges, including:

- Real-time processing complexity: Real-time processing of healthcare data, such as continuous patient monitoring or real-time diagnostic systems, is more complex and resource-intensive than batch processing. Sophisticated algorithms and high computational power are needed to handle the continuous flow of data and provide timely insights (Porambage et al. 2018);
- Scalability issues: Handling large volumes of continuous data flow, such as patient vital signs or telemetry data, presents significant scalability challenges. Efficient data management and processing techniques are essential to ensure smooth operation and avoid bottlenecks (Kumar et al. 2023);
- Latency concerns: Achieving low-latency responses is crucial in healthcare applications to enable timely interventions, such as alerting medical personnel about critical changes in patient status. Optimizing data pipelines and utilizing efficient algorithms are key to minimizing delays and ensuring prompt action (Mehmood and Anees 2020);
- Data quality and consistency: Maintaining data quality and consistency in a streaming context is critical due to the rapid nature of data arrival. Real-time validation and cleaning processes are essential to ensure the accuracy and reliability of the data (Gorawski et al. 2023); and
- Integration with existing systems: The integration of streaming analytics with existing healthcare systems, such as EHRs, further complicates the landscape. Seamless and secure data exchange mechanisms are necessary to ensure data accessibility and interoperability (Ahmed et al. 2023).

IoMT devices have many advantages for patients and healthcare professionals, but several drawbacks must be resolved (Manogaran et al. 2018). The most important constraints that IoMT faces include restricted accessibility, technical complexity, a lack of clinical evidence, ethical concerns, regulatory obstacles, and integration with current systems (Vishnu et al. 2020). To ensure that IoMT devices are used safely, effectively, and ethically, healthcare practitioners, manufacturers, and governments must work together to address these limitations.

6.5 Future development

Healthcare providers, tech companies, legislators, and patients will need to work together to overcome the complex and multidimensional issues and constraints of the IoMT. Some potential plans and objectives for overcoming the difficulties and restrictions of the IoMT devices are as follows:

- **Increasing accessibility**: Programs that offer financial support to patients who cannot afford these devices are part of improving accessibility to IoMT devices. Efforts to increase internet connectivity in remote and under-served locations can also help patients have easier access to IoMT devices (Al-Jaroodi et al. 2020).
- **Simplifying user interfaces and training**: By streamlining user interfaces and training, IoMT devices can be used more easily by patients and medical professionals. Technical complexity can be reduced by creating simple, intuitive user interfaces, and training programs can guarantee that patients and healthcare professionals are competent in us-

ing IoMT devices (Vishnu et al. 2020).

- **Boosting clinical evidence**: To address the scant clinical evidence available for IoMT devices, more thorough and extensive studies that assess their efficacy and safety are required. These studies can offer the data necessary for healthcare professionals, payers, and legislators to make well-informed decisions regarding using IoMT devices (Al-Jaroodi et al. 2020).
- **Resolving ethical issues**: Manufacturers and healthcare providers can put rigorous privacy and security measures in place to solve ethical issues with IoMT devices. Patients should be made aware of who will access their data and how it will be utilized. Patients, healthcare providers, and manufacturers can all benefit from clear and open policies regarding data security and privacy (Babar and Rahman 2021).
- **Uniforming regulatory framework**: This standardized framework is required to guarantee the safety and effectiveness of IoMT devices and overcome regulatory concerns. The regulatory process should be streamlined, redundant tasks should be eliminated, and innovation should be encouraged while maintaining patient safety (Babar and Rahman 2021).
- **Integrating with current systems**: To ensure that patient data can be shared effectively and efficiently, improvements in integration between IoMT devices and current healthcare systems are required. By creating standardized protocols and data formats for IoMT devices and healthcare systems, this integration can be accomplished (Al-Jaroodi et al. 2020).

To guarantee that IoMT devices are utilized safely, effectively, and ethically to enhance patient outcomes, these previously mentioned initiatives require collaboration between healthcare practitioners, technology manufacturers, legislators, and patients (Zhou et al. 2020).

Healthcare delivery has already undergone a considerable transformation thanks to the IoMT, but this technology is anticipated to have even greater effects. The following are a few of the most significant future directions for IoMT:

- AI and ML into IoMT devices may increase the precision and speed of data analysis, resulting in more accurate diagnoses and successful therapies. Moreover, AI and ML systems can suggest individualized patient treatment options and forecast future health outcomes (Baali et al. 2017).
- Wearable IoMT devices, such as smartwatches and fitness trackers, are commonplace, but more innovation is anticipated in this area. Future wearable technology is expected to have more sophisticated sensors and capabilities, enabling continuous monitoring of vital signs and other health indicators. Moreover, wearable IoMT technology may be very helpful in managing and preventing chronic illnesses (Zhou et al. 2020).
- IoMT devices are frequently used with telemedicine, enabling remote medical consultations and treatment. Future advances in telemedicine might result in even more sophisticated virtual consultations, where medical professionals might remotely check on patients' vital signs and other health indicators and modify treatment regimens as necessary (Baali et al. 2017).
- 5 G networks and edge computing might completely transform IoMT devices by enabling quicker and more dependable data flow and analysis. These technologies might

also make it possible to create IoMT devices that are more sophisticated and complex, including autonomous diagnostic tools and surgical robots (Baali et al. 2017).
- The security and privacy of patient data in IoMT devices may be enhanced using blockchain technology. The usage of blockchain can establish a decentralized, open, and transparent system for storing and exchanging patient data, ensuring that the information is safe and only available to those with the proper authorization (Khan and Algarni 2020).
- Healthcare streaming data analytics can be enhanced by developing more robust and scalable frameworks for improving healthcare streaming data analytics, enhancing real-time processing capabilities, and improving integration with existing healthcare systems (Ahmed et al. 2023). The research aims to create algorithms that handle large volumes of data with minimal latency and high accuracy for real-time clinical decision-making (El-Ganainy et al. 2020). Specialized tools and frameworks tailored to healthcare needs will provide efficient solutions for continuous patient monitoring and real-time diagnostics (Palanisamy and Thirunavukarasu 2019). Better integration with electronic health records will enhance patient care and operational efficiency. Additionally, using ML and AI will advance predictive analytics for early disease detection and personalized treatment plans (Cerchione et al. 2023).

7 Conclusion

IoMT has the potential to completely transform the healthcare sector by offering real-time data analysis, remote monitoring, and individualized treatment alternatives. Many technologies, including wearables, blockchain, and AI, can improve the effectiveness and efficiency of IoMT solutions. IoMT has advantages but drawbacks, including regulatory compliance, interoperability problems, and worries about data security and privacy. Despite these difficulties, the IoMT is altering how healthcare is provided and promises to enhance patient outcomes while significantly cutting costs. Stakeholders must cooperate to address issues and advance the appropriate and ethical use of these technologies as the IoMT develops new technologies and continues to change. Overall, the IoMT is a promising field that can assist patients and healthcare professionals significantly and leads to major improvements in patient outcomes and healthcare delivery. It should be encouraged to continue developing and implementing.

Acknowledgements Not applicable.

Author Contributions Conceptualization, methodology, statistical analysis, data analysis: SEE, ABE, and KMS; literature review, discussion, writing-original draft preparation, data downloading: SEE, ABE, and AAA; writing-review and editing: KMS, AAA, and ABE; visualization: SEE, KMS and AAA; supervision: KMS. All authors read and approved the final manuscript.

Funding Open access funding provided by The Science, Technology & Innovation Funding Authority (STDF) in cooperation with The Egyptian Knowledge Bank (EKB).
No funds are available for this research.

Data availability The software and code developed for this study are available upon request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval Not applicable.

Consent for publication Not applicable.

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